

WAGE FLEXIBILITY UNDER SECTORAL BARGAINING

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Abstract

Sectoral contracts in many European countries set wage floors for different occupation groups. In addition, employers often pay a wage premium (or *wage cushion*) to individual workers. We use administrative data from Portugal, linked to collective bargaining agreements, to study the interactions between wage floors and wage cushions and quantify the impact of sectoral wage floors. Although wages exhibit a “spike” at the wage floor, a typical worker receives a 20% premium over the floor, with larger cushions for older and better-educated workers and at higher-productivity firms. Cushions also allow wages to covary with firm-specific productivity, even within sectoral agreements. Contract negotiations tend to raise all wage floors proportionally, with increases that reflect average productivity growth among covered firms. As floors rise, however, cushions are compressed, leading to an average passthrough rate of about 50%. Finally, we use a series of counterfactual simulations to show that real wage reductions during the recent financial crisis arose through reductions in real wage floors, reductions in real cushions, and a re-allocation of workers to lower wage floors. Offsetting these effects was a rapid rise in education of new cohorts, which in the absence of other factors would have led to rising real wages. (JEL: J31, J41, J51)

1. Introduction

How does collective bargaining affect wages? Much of the existing research on this issue focuses on the U.S. (e.g., Freeman and Medoff, 1984; Lewis, 1986; Farber et al., 2021), where union contracts set wages for *jobs*. In this setting an increase in negotiated wage rates translates directly to an increase in wages for workers who remain in the same job. Collective bargaining agreements in many European countries work differently: these agreements specify a set of *wage floors* for different occupation

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groups. Employers can (and often do) pay idiosyncratic wage premiums on top of the floors.¹ These premiums—which Cardoso and Portugal (2005) labeled “wage cushions”—partly undo the wage-standardizing features of U.S.-style collective bargaining, contributing to within- and between-group pay inequality.² Premiums can also adjust when floors change (or are frozen), providing a degree of wage flexibility that is absent in the U.S. setting.³

In this paper we explore the relationship between collectively bargained wage floors and actual wages in Portugal, using individual wage records linked to collective bargaining agreements from 2008 to 2016. The Portuguese system of sectoral bargaining is broadly similar to the systems in Spain, Italy, Belgium, Netherlands, and France (see Schulten, 2016). Moreover, the prevalence of pay rates in excess of negotiated wage floors parallels the situation in other countries. Thus, we believe there are general lessons to be drawn from a study of Portugal. The setting is also interesting because as part of a 2011 debt relief package, a *Troika* of international agencies pushed for legislative changes that would reduce the coverage of sectoral bargains.⁴ This effort largely failed. Nevertheless, as we will show, significant downward real wage adjustments occurred within the framework of the existing bargaining system.

The key to our analysis is the ability to link individual workers in the annual census of employees in Portugal—known as *Quadros de Pessoal* (QP)—to the collective bargaining agreements (CBA) and wage floors that apply to their jobs. This is made possible by two institutional facts. First, the QP identifies the CBA for each worker covered by a union contract, as well as a job title that in principle specifies their wage floor. Second, information on all newly negotiated CBA’s, including tables of wage floors for different occupational groups, is published by the Ministry of Labor. Setting aside difficulties in matching, it is therefore possible to assign wage floors to covered workers observed in the QP in October of each year. While many previous studies have attempted to link *subsets* of workers to their associated wage floors (e.g., Cardoso and Portugal (2005) for Portugal; Card, Devicienti and Maida (2014) for Italy; Deelen and Euwals (2014) for Netherlands; Diez-Catalan and Villanueva (2014) for Spain) we

1. See Holden (1989, 1998) for the case of Norway; Calmfors and Nymoen (1990) for a broader discussion of the Nordic countries; Hübler and Jirjahn (2003) and Jung and Schnabel (2011) for the case of Germany; Ordine (1995) for Italy; Dolado et al. (1997) for Spain; Butter and Eppink (2003) for The Netherlands, and Cardoso and Portugal (2005) and Bastos et al. (2009) for earlier analyses of Portugal.

2. Freeman (1980) noted that the variance of wages is lower in the union sector than the non-union sector and credited this in part to the elimination of idiosyncratic wage variation within jobs. Similarly, Ashenfelter (1972) noted that unions raise wages of black workers relative to whites and suggested that this arose in part because of standardization policies that reduce racial wage gaps within jobs. See Card, Lemieux and Riddell (2004) for more discussion.

3. This fact is widely recognized in the literature on “wage drift”—Phelps Brown (1962) presents an early, informative analysis. See also Calmfors (1993) and Schlicht (1992).

4. See Blanchard et al. (2014) for a discussion of the IMF’s recommendations, which appear to have been adopted in the Memorandum of Understanding between the Government of Portugal and the European Commission (EC), European Central Bank (ECB), and International Monetary Fund (IMF).

believe this is the most comprehensive panel data set assembled to date that combines information on collectively bargained wage floors and actual wages.⁵

Two initial questions, highlighted by the goals of the Troika, are: How did the share of workers covered by CBA's change between 2008 and 2016? And how do uncovered workers compare to covered workers? Consistent with other recent studies (e.g., Addison, Portugal, and Vilares, 2017) we show that the fraction of full-time workers in QP covered by CBA's fell only slightly, from 90% in 2008 to 87% in 2016. We also show that uncovered workers in Portugal earn significantly *higher* wages than covered workers, contrary to the situation in countries such as the U.S. or U.K.

We then present a descriptive analysis of the role of wage floors in between- and within-group wage variation. We show that the log of an individual's total monthly wage can be decomposed into four components: (i) the minimum wage; (ii) the worker's *relative wage floor* (i.e. the floor relative to the minimum wage); (iii) the gap between the base wage and the wage floor (i.e., the wage cushion); and (iv) regular supplementary payments (including meal subsidies and shift premiums).⁶ As was documented by Cardoso and Portugal (2005) using QP data for 1999, we find that differences in relative wage floors and differences in mean wage cushions both contribute to inter-group wage differences. For example, about 30% of the wage gap between men and women is attributable to higher wage floors for men, and 60% to higher mean wage cushions for men. We also show that wage inequality *within* skill groups reflects variation in both wage floors and wage cushions, as well as the covariance between them.

Within a given sectoral agreement firms have some latitude in assigning workers to different floor categories, and even more latitude in determining wage cushions. Both factors contribute to the cross-sectional variation in wages *within CBA's*. Classifying firms into deciles of average value added per worker, we show that mean base wages at top decile firms are over 40 log points higher than mean base wages at bottom decile firms *in the same sectoral agreement*. Around 10% of this effect is attributable the assignment of workers to higher wage floors at top decile firms, while 90% is attributable to higher wage cushions. Thus, wage cushions play a particularly large role in within-CBA wage flexibility.

Next, we study the renegotiation process for wage floors. We show that all the floors in a given CBA adjust by virtually the same percentage when the contract is renegotiated. We then relate this average floor adjustment factor to measures of productivity growth among firms covered by the contract. We focus on two closely related questions: (1) Are wage outcomes driven by *average* productivity growth of covered firms, or by the high- or low-performers in the covered set? (2) How sensitive

5. Gautier et al. (2021) have a rich data base that combines individual wages and collectively bargained wage floors for France, but cannot directly link workers to specific wage floors within the relevant agreement. Fougère et al. (2018) use similar data on wage floors to study the adjustment of floors to inflation and changes in the minimum wage.

6. Meal allowances are widespread in Portugal, in part because they are tax exempt up to a fairly generous level (currently up to 7.63 Euros per day).

are negotiated wage floors to productivity growth of covered firms? We find that floor adjustments respond to the central tendency of growth in value added per worker among covered firms, rather than to upper- or lower-tail growth, with an elasticity of around 0.10 —as big or bigger than the typical elasticities estimated in the micro rent sharing literature (see Card et al., 2018).

The net effect of wage floor adjustments depends on how wage cushions respond to these adjustments. If employers react to keep the same wage cushion as a floor is raised, then wage floor increases will pass through fully to actual wages. If cushions are compressed as floors rise, however, the passthrough rate will be lower. To estimate passthrough rates we calculate the change in base wages that would occur if each worker maintained the same gap between their wage and the wage floor as floors are changed. We then regress actual wage increases on these simulated increases, using both OLS and an instrumental variables approach that takes the average simulated increase in wages for all workers at the same firm as an instrument for the worker-specific effect of the floor increase.

We find that the average passthrough rate of floor increases is around 50%, with a higher passthrough rate for workers with smaller wage cushions. This pattern is similar to the spillover effect of a minimum wage increase (e.g., Cengiz et al., 2019; Fortin, Lemieux and Lloyd, 2021), though we find that the impact of wage floor increases extends further up the distribution. We also test for but find no evidence of asymmetry of responses to real wage floor changes arising from new contract negotiations versus those attributable to inflation.

We also examine the effect of wage floor adjustments on employment. Specifically, using the same instrumental variable we use to study the passthrough of floors to individual wages, we relate firm-wide employment changes to the simulated increase in base wages of its employees caused by changes in wage floors. Our estimates suggest that employment is largely unaffected by higher wage floors, though we cannot reject small negative impacts.

In the final section of the paper we conduct a simulation analysis to understand how changes in wage floors and wage cushions, as well as movements of workers between jobs with different wage floor categories, contributed to the adjustment of real wages between 2010 and 2016, as Portugal suffered through a prolonged recession. We begin by computing mean real wages for workers in different gender-education-age groups in 2010. We then increment all wage floors to incorporate renegotiations between 2010 and 2016, but keep each worker in the same floor category and hold constant their wage cushions and supplementary payments. The comparison between this counterfactual and the 2010 baseline summarizes the net effect of wage floor adjustments, and shows a 2.5 ppt *reduction* in average real wages attributable to the erosion in real floors over the 6 years. Next, we reweight skill groups in 2010 to their 2016 shares to measure the effects of demographic change. Driven by a rapid rise in shares of better-educated workers, this yields a 7.4 ppt *increase* in mean real wages in the economy as a whole that would have occurred if wage floors, wage cushions, and the assignment of workers to floors had remained constant.

We then consider a counterfactual based on workers observed in 2016, using their actual wage floors as of 2016 but simulating the wage cushion each worker would have earned in 2010 (by drawing from the distribution of cushions in 2010). Relative to the previous simulation, this counterfactual reveals the net effect of the reallocation of workers across wage floor groups that occurred between 2010 and 2016, and yields a 4.8 ppt *reduction* in mean real wages for workers as a whole. Finally, we give each worker their actual wage cushion in 2016 (rather than a simulated 2010 wage cushion). This final step shows that changes in wage cushions within wage floor categories led to a further 2.5 ppt *reduction* in mean real wages.

Despite concerns that sectoral bargaining limits the responsiveness of real wages to negative shocks, our simulations suggest that real wages fell substantially during the debt crisis. The declines were particularly large for university-educated workers, whose mean real wages fell by 16 ppt between 2010 and 2016, reflecting a combination of declining real wage floors (-4.4 ppt), declining real cushions (-6.2 ppt), and a reallocation of jobs toward lower wage floor categories (-8.4 ppt). Real wage cuts for lower-paid workers were smaller but still significant: young high school-educated females and males, for example, experienced declines of 4.8 ppt and 5.6 ppt, respectively.

Our findings contribute to three separate strands of research. First, we contribute to a macro-oriented literature that compares different collective bargaining systems (e.g., Calmfors and Driffill, 1988; Calmfors, 1993; Nickell and Layard, 1999). This literature often assumes that sectoral agreements set wages for covered firms, ignoring employer-determined wage cushions—a simplification that overstates the rigidity of Portuguese wage setting.

Second, we contribute to the micro-oriented literature linking union-wage setting to wage inequality (Freeman, 1980; Card, 1992; DiNardo, Fortin and Lemieux, 1996; Farber et al., 2021). Building on Cardoso and Portugal (2005), we show that in a European setting, idiosyncratic wage premiums are important determinants of within-group and between-group inequality. The size and distribution of these premiums helps explain why, despite high CBA coverage rates, Portugal also has relatively high wage inequality.

Finally, we contribute to the “micro Phillips curve” literature (e.g., Riddell, 1979; Card, 1990; Christofides and Oswald, 1992) that examines the determinants of negotiated wage outcomes using union contract data. Our data allow us to examine the full set of wage floors within a contract, rather than just the “base wage” for lower skilled workers that is usually analyzed in this literature. We also study how multi-employer agreements are impacted by the distribution of firm-specific productivity growth among covered firms. Finally, we show how collectively bargained wage floors affect individual wage outcomes as well as within- and between-group wage inequality.

Níveis	Grupo A	Grupo B
XII	960,0 €	930,0 €
XI	895,0 €	887,0 €
X	770,0 €	735,0 €
IX	700,0 €	670,0 €
VIII	630,0 €	610,0 €
VII	585,0 €	575,0 €
VI	540,0 €	540,0 €
V	532,0 €	532,0 €
IV	531,0 €	531,0 €
III	530,0 €	530,0 €
II	450,0 €	450,0 €
I	440,0 €	440,0 €

FIGURE 1. Example of wage table from BTE. “Contrato coletivo entre a Associação da Hotelaria, Restauração e Similares de Portugal (AHRESP) e o Sindicato dos Trabalhadores e Técnicos de Serviços - SITESE - Alteração salarial e outras”.

2. Setting and Conceptual Framework

Sectoral Bargaining in Portugal and Reforms During the Debt Crisis. In the system established in Portugal in the 1970s and still in place today, employer associations representing firms in a particular industry (and in some cases region) sign CBA's with one or more trade unions.⁷ Although these agreements technically cover only union members, in practice employers extend the agreements to their entire workforce, regardless of membership status.⁸ Under the laws and practices that were largely in place in 2010, the bargaining parties would often file a request with the Directorate-General for Employment and Labor Relations to extend the agreement to other firms in the same sector—a request that was normally granted (see Naumann, 2018). Contract provisions could also be voluntarily adopted by employers in the industry.

Each CBA contains a variety of clauses prescribing work rules and practices, as well as a set of wage floors that prevail during the term of the contract. Figure 1 presents an example of the table of wage floors from a typical agreement—in this case

7. There are two main union confederations in Portugal - the União Geral de Trabalhadores (UGT) and the more radical Confederação Geral dos Trabalhadores Portugueses (CGTP). Often, an employer association will sign separate *but identical* agreements with different unions—typically, one affiliated with UGT and another affiliated with CGTP. In our analysis below we consolidate such duplicate agreements and treat such parallel agreements as a single one.

8. We verified this directly by looking at the distribution of the fraction of employees within each firm classified as covered by a CBA in the QP. This distribution is effectively comprised of a mass at 100% and a smaller mass at 0%.

a 2016 agreement between the Association of Hotel and Restaurant Employers and the Union of Service Workers. This wage table distinguishes between two subgroups of employers (groups A and B) and 12 different wage floors, ranging from 440 to 960 Euros per month.⁹

Collectively bargained wage clauses almost always have a nominal duration of one year. In case a new agreement has not been negotiated, however, the old agreement remains in force, and in the early years of our sample (2008-2009) a typical new agreement was updating a contract that was negotiated about two years earlier (see Section 3.1, below). Prior to 2003 the Labor Code required that any new agreement be at least as favorable to workers as the old agreement and also prevented firms from withdrawing from a CBA. These rules were relaxed by amendments in 2003 and 2009 that allowed new agreements to loosen work rules and lower wage floors. The 2003 and 2009 amendments also created a process for CBA's to expire, though procedures governing the granting of extensions were unchanged.¹⁰

At the peak of the financial crises in 2011, the Portuguese government signed a memorandum of understanding (MOU) with the European Commission (EC), European Central Bank (ECB) and International Monetary Fund (IMF)—the so-called *Troika*—committing to a wide range of policy reforms, including revisions of the contract extension framework that were intended to reduce the coverage of sectoral agreements and encourage firm-level bargaining.¹¹ Ultimately these reforms ran into legal challenges, as well as opposition from employer associations, many of which supported the existing extension framework (see Naumann, 2018). After Portugal exited the financial aid program in 2014, the new center-left government adopted a series of revisions that more or less restored the pre-crisis bargaining framework.

Wage Setting Under Sectoral Bargaining. The wage floors in Portuguese CBA's set a lower bound on basic pay for workers in each occupational category. As in other European countries, however, firms can and do offer many workers a wage that is higher than the minimum for their category. This differs from the typical situation in the U.S., where a union contract specifies a grid of wages for different jobs, and *all workers in the same job receive the same pay*—a wage standardizing property that is arguably

9. These are monthly salaries for full time workers, net of payroll taxes. By law, workers receive 14 monthly salaries. As of 2016 (for which the floors apply) the minimum wage was 530 Euros, so group III has a floor at the national minimum wage. The two bottom groups are apprentices, who face a minimum of 80% of the regular minimum wage.

10. The number of collective bargaining agreements that were determined to have expired under these rules is low: a total of 15 expiration notices were published in 2009; and over the period from 2010 to 2016 another 17 expiration notices were published (Portugal, CRL, 2020: 58; Portugal, MTSS, 2016: 374). The number of agreements that actually expired was somewhat smaller because of subsequent Court decisions.

11. A key goal for the Troika was to reduce the coverage of sectoral agreements and encourage firm-specific agreements negotiated by works councils (see European Commission, 2011, p. 54). The Troika agreement ignored the fact that the Portuguese Constitution gives trade unions the exclusive right to bargain for workers. Two other practical problems were that there were less than 200 works councils in the entire country (MTSS, 2006), and that the vast majority of Portuguese firms are very small and have limited capacity for bargaining on their own.

a defining feature of unionized wage setting in a U.S.-style system (Ashenfelter, 1972; Freeman, 1980). In addition, most workers in Portugal receive regular “supplementary” payments, including tax-free meal subsidies, that are the same from month to month and may be impacted by collective negotiations.

To clarify the role of these various components, let W_{it} represent the net monthly base wage for worker i in year t and let $F_{i,t}$ represent the wage floor that applies to that worker. Let $H_{it} = W_{it} - F_{it}$ represent the absolute gap between the base wage and the wage floor, and let S_{it} represent the regular monthly supplemental payments received by worker i in year t .¹² Then we can decompose the base monthly wage and the corresponding total monthly wage (W_{it}^T):

$$\begin{aligned} W_{it} &= F_{it} + H_{it} \\ W_{it}^T &= F_{it} + H_{it} + S_{it} \end{aligned}$$

For most of our analysis below we work with logarithms of wages rather than levels. Letting $w_{it} \equiv \ln W_{it}$ represent the log of the monthly base wage, and $w_{it}^T \equiv \ln W_{it}^T$ represent the log of the monthly total wage, we can write:

$$w_{it}^T = f_{it} + h_{it} + s_{it} \quad (1)$$

where $f_{it} \equiv \ln F_{it}$ is the log of the wage floor for worker i in year t ,

$$h_{it} \equiv \ln \frac{W_{it}}{F_{it}}$$

is the proportional wage premium received by the worker over his or her wage floor (which we refer to as the worker’s “wage cushion”), and

$$s_{it} \equiv \ln \frac{W_{it} + S_{it}}{W_{it}}$$

represents his or her regular supplementary payments, expressed as a share of the base wage.

In the presence of a national minimum wage, it is helpful to decompose the log wage floor into the sum of the log of the minimum wage ($m_t \equiv \ln M_t$) and the gap between the floor and the minimum wage:

$$f_{it} = m_t + rf_{it}$$

where rf_{it} is the wage floor relative to the minimum wage:

$$rf_{it} = \ln \frac{F_{it}}{M_t}.$$

12. In Portugal (as elsewhere in Continental Europe) wages are normally expressed as monthly full-time rates, net of any employee payroll taxes. Wage floors in CBA’s and the national minimum wage are similarly expressed. Moreover, workers receive 14 monthly salaries per year.

Substituting into equation (1) we get a simple four component model of log wages:

$$w_{it}^T = m_t + rf_{it} + h_{it} + s_{it} \quad (2)$$

that expresses the log total wage for individual i in year t as the **sum** of the minimum wage, the relative wage floor for the worker's job, her wage cushion, and her regular supplementary payments. This additive structure is very convenient for decomposing the variance of the log of total wages (see Section 4, below), for addressing the causal question of how actual wages respond to adjustments in wage floors (see Section 6), and for considering counterfactual scenarios, such as one in which floors are raised and all wage cushions remain constant, so each worker maintains a fixed (proportional) pay premium over his or her floor (see Section 8).

3. Assigning Wage Floors to Workers

In this section we describe our data base of workers with assigned wage floors. We begin with an overview of our data base of CBA's. We then discuss the Quadros de Pessoal (QP) and our procedure for assigning wage floors to workers in QP.

3.1. Data on Collective Agreements – BTE

All newly negotiated CBA's in Portugal are published in the Labor Bulletin (*Boletim do Trabalho e Emprego*, BTE) and are available in an online archive (<http://bte.gep.msess.gov.pt>). We began our data assembly process by extracting information for agreements published between 2008 to 2016 that included a salary clause or wage table. For each agreement we extracted:

- the names of the union(s), employer association(s) and other information that formally identifies the contract
- the type of agreement (sectoral agreement, company agreement, multi-company agreement, government directive)¹³
- the starting date; expiration date; and reference information on the preceding agreement.

We also collected information on the categories and wage floors in the wage tables. The system for designating floor categories varies widely across contracts but in most cases we are able to devise a list of job titles/occupations included in each category, and construct a longitudinal data base of wage floors for each CBA and floor category.

There are a number of issues that have to be addressed in constructing an accurate panel of wage floors. One is that we only observe wage floors when a contract is

13. Multi-company agreements (*acordo coletivo*) are legally distinct from sectoral agreements (*contrato coletivo*) and are particularly common in the finance and utility sectors. Government directives are mandated agreements imposed in the absence of any other collective agreement (*portaria de condições de trabalho*) or in case of an unresolved dispute (*decisão arbitral*).

actually updated. Thus, the first observation for each CBA/floor category occurs at the time of the first contract renegotiation after January 1, 2008. A second issue is that floor increases are sometimes back-dated. Since our interest is in the effect of wage floors on the current (flow) cost of labor, we measure the *prevailing* wage floor as of October of each year (the reference date of the QP survey), ignoring any back payments awarded by subsequent agreements. A third issue is that increases in the national minimum wage can over-ride wage floors for lower-paid workers, particularly if the contract has not been renegotiated recently. In accordance with the labor law, we update all wage floors to meet the minimum wage as of the reference date of the QP. A fourth complication is that some agreements (such as the one underlying the wage table in Figure 1) specify separate wage floors for subgroups of firms (e.g., based on revenues), or workers (e.g., based on tenure). We keep track of the subgroup classification system and attempt to assign the correct floor to a worker, though that is not always possible.

A final issue is that an employer or employer association will often sign separate *but identical* agreements with different unions—typically, one affiliated with UGT and another affiliated with CGTP. We consolidate such duplicate agreements, reducing the total number of agreements over the 2008–2016 period from 1,467 to 1,061 (See Online Appendix A and Online Appendix Table A.1). We also drop agreements covering firms in agriculture or fisheries, or those in Madeira or the Azores. We are left with 988 new consolidated agreements that form our basic CBA data set. Around 50% of these are sector-wide contracts, just over 10% are multi-company contracts, and the remaining 38% are CBA's covering a single firm (see Online Appendix Tables A.2 and A.3).

Column 1 of Table 1 shows the number of unduplicated new agreements in our basic CBA data set by year of renegotiation, while column 2 shows the share of those agreements that were sectoral CBA's.¹⁴ Close to 200 (consolidated) agreements were reached in 2008. The number then began to fall off, reflecting the tendency for renegotiations to slow down in the face of worsening economic conditions. In 2012 and 2013 the number was particularly low, driven by the severe recession and uncertainty over collective bargaining institutions in the aftermath of the MOU with the Troika. Following the nascent recovery and legislative changes in 2014 that re-established the framework for contract extensions, the number of new agreements rose to around 90 per year in 2014–2016.

Although nearly all collective bargaining agreements in Portugal (97% in our sample) have a nominal one-year duration, an existing CBA remains in force until a new one is negotiated (or in very rare cases when an employer exits the agreement). As shown in column 3 of Table 1, in 2008 the mean elapsed time since the publication date of the previous agreement was 20 months—implying a delay of about 8 months between the expiration of the old contract and the publication date of the new one. By 2015 the time since last agreement had risen to 37 months, implying a delay of over

14. We emphasize that the numbers of agreements shown represent counts after consolidating duplicated agreements. The numbers of agreements prior to this adjustment are shown in Online Appendix Table A.3. A typical sectoral agreement covers firms in multiple regions: weighting by employment, 86% of sectoral agreements include workers in all 5 NUTS2 regions of Portugal.

2 years between the expiration date and the renegotiation date. The increase in delay time was particularly pronounced for sectoral contracts, driven by the near-collapse in renegotiation of these agreements in 2012 and 2013. As a consequence of these long delays, by 2014 many workers were covered by floors that were 2-3 years old, a situation that was only partly remedied by the upswing in negotiations in 2015 and 2016.

3.2. *Quadros de Pessoal*

Quadros de Pessoal (QP) is an annual census of employers conducted by the Ministry of Employment. Firms with at least one wage earner are required to submit their full roster of employees as of the reference week in October, as well as a variety of other information (including annual sales). They are also required to post their employee roster (with names, job titles, and monthly pay) inside their premises, reducing the likelihood of misreporting or under-reporting. The Ministry distributes an electronic version of the data set that has longitudinal identifiers for each firm and each worker. We use QP data for the period from 2008 to 2016.

The data for each worker include gender, age, education, occupation, date of hire, nationality, monthly earnings (split into several components), hours of work (normal and overtime), as well as the name of the CBA that the worker is covered by (if any). Unfortunately, the QP does *not* report the actual wage floor for the worker or the name of the floor category as used in the BTE. Instead, it reports a *job title* or *professional category* of the worker, which in many cases can be matched to the list of job titles or occupations reported for the floor categories in BTE.

In addition to the information collected by the QP itself, we also have access to matched income statement/balance sheet information for most employers, linked to the QP by the National Statistical Office.¹⁵ Relevantly for our purpose, these business statistics report the yearly Gross Value Added at the firm level. We do not construct this variable. It is directly supplied by the firm. It is defined as the production value minus the purchase of goods and services, after all production taxes have been paid and all production subsidies have been received (GVA at ‘factor cost’ or ‘basic prices’), i.e. what remains to be distributed among the production factors.

Starting from the universe of observations in QP we exclude workers under the age of 18 or over 64, those in Madeira and the Azores, and those employed in agriculture and fisheries (see Online Appendix A). We also exclude apprentices (3.5% of the relevant sample), workers who are not employed full time (15.1%), and those with missing information on wages (8.9%, including unpaid family members and firm owners) or education/date of hire (0.1%). Columns 4-6 of Table 1 report the resulting number of workers in our QP sample each year, the fraction that are reported as covered by a CBA, and the fraction covered by sectoral agreements. On average we have

15. The Integrated Business Accounts System – IBAS (Sistema de Contas Integradas das Empresas – SCIE) covers the non-financial business sector. The linked QP data are distributed in an anonymized format.

about 1.85 million workers per year, with a dip during the most severe recessionary years and a partial recovery by 2016. The collective bargaining coverage rate starts at 90% in 2008, remains relatively steady until 2011, then declines slightly each year thereafter, ending at 87%. On average 81% of covered workers are covered by a sectoral agreement, a fraction that fell slightly over our sample period, from 83% to 80%.

In Online Appendix B we use a simple dynamic model to decompose the year-to-year changes in collective bargaining coverage in our QP sample, focusing on worker-level transitions between three states: employed and covered by a CBA, employed and not covered by a CBA, and not employed (i.e., not included in QP in a given year). There are no major changes over time in the probability that people retain their coverage status, or in transition rates into or out of jobs covered by a CBA (see Online Appendix Table B.1). There was a slowdown in the probability that people entered the workforce in 2011–2014, and a slight reduction in the fraction of new entrants starting a job covered by a CBA. Together these factors account for most of the (relatively modest) losses in coverage after 2011.

3.3. Assigning Wage Floors to Workers in QP

We used a two-step process to assign wage floors to workers in QP. We first matched contracts in QP to those in our BTE database. We then attempted to match the wage floor groups within a contract in BTE to the job category codes reported in QP.

The matching of contracts was done by hand since the CBA names in the two data sources can differ and the QP often uses outdated names. Broadly, the steps included: inspection of the text of each agreement to identify likely matches; construction of consistent longitudinal information on the renegotiation dates of each agreement and on the reported numbers of covered workers and firms to confirm matches; inspection of longitudinal information on workers in QP to identify likely CBA name changes; searches on the web pages of trade unions or employer associations; and telephone or email contacts with trade unions. With these steps we were able to match nearly all contracts mentioned in QP to an agreement in BTE.

Matching of the wage floors in BTE to the worker categories in QP was also done by hand, and was more difficult. We began by inspecting the text of each agreement in BTE to find a list of all jobs/job titles in each floor group.¹⁶ Next, we matched the BTE *floor groups* in a given CBA to the QP worker *categories* for the same CBA, again by direct inspection of the possible $m \rightarrow n$ matches for each CBA. In agreements setting different wage floors for workers depending on their date of hire, tenure, or skill, we attempted to use information in QP to assign workers to the correct wage floors. Likewise, whenever the applicable wage floor depended on firm attributes reported in QP, such as the firm's industry or employment, we matched the worker category in

16. We often had to track past agreements to find the full list of job titles in each floor group because in some cases —such as that shown in Figure 1— only a group code is reported.

QP to its wage floor accordingly. Online Appendix A provides further details on the process of matching.

Despite our best efforts we were only able to match slightly over half the workers in QP covered by a CBA to their wage floor (see Online Appendix Table A.2). The main obstacles were (1) lack of information on the variables needed to assign workers to specific floors within a CBA; (2) too many sub-floors for each occupational category; (3) lack of obvious matches between the occupations or job types specified in BTE and the job titles used in QP. Columns 7-9 of Table 1 present some information on the subset of workers in QP that were successfully assigned a floor. The fraction of matched workers rises from 32% in 2008 to 44% in 2010 and is more or less stable thereafter. These shares refer to the full QP dataset reported in Column (4). If, instead, we consider only those workers actually covered by collective bargaining with a job category reported, the shares with assigned floors rise to 39% in 2008, 52% in 2010, and 53% over the whole period under analysis. The lower rate at the start of our sample is due to the fact that many workers in QP in October 2008 or 2009 were covered by floors that were last renegotiated in 2006 or 2007, prior to the start of our BTE database. By the time of the 2010 QP most workers in QP were covered by an agreement that was updated between January 2008 and September 2010. We note that in a typical year after 2010 our matched database includes about 2,500 separate wage floors.¹⁷

3.4. Comparisons of Workers by Coverage and Floor Assignment Status

Before proceeding with an analysis based on the subset of workers with matched floors we examine two questions: How do covered workers with a matched floor compare to those for whom we were unable to assign a floor? And how do workers who are uncovered by CBA's compare to covered workers?

Table 2 presents some simple data that address these questions: we show characteristics and wage outcomes for all workers, for those who are covered and uncovered by a CBA, and for covered workers with and without a matched wage floor. Focusing first on the data in columns 4 and 5, we conclude that covered workers who can be assigned a wage floor are broadly similar to those who cannot. In particular their gender, education, experience, job tenure and mean log wages are quite similar. Importantly, this similarity is also true year-by-year (see Online Appendix Table C.1), suggesting that we can draw broader conclusions from an analysis of data for workers with assigned wage floors.

On the other hand, comparisons between columns 2 and 3 show that workers with and without CBA coverage are substantially different. Uncovered workers are much more likely to have a university-level education (38% versus about 17% for covered

17. Martins (2021) claims that there are 30,000 minimum wage floors in Portugal. His analysis counts all job categories within the CBA's identified in QP, without taking into account the duplication of CBA's or the fact that on average a wage floor group in BTE actually incorporates roughly 4 job categories in QP. Together these corrections imply that there are only about 5,000 separate wage floors at any point in time, roughly half of which we are able to match to a wage floor published in BTE.

workers), have somewhat fewer years of experience and job tenure than covered workers, and have about 20% higher wages. This wage advantage appears to be driven by worker skills: when we fit a model of wages that includes the coverage status of the current job and worker fixed effects it falls to essentially 0. Not surprisingly, wages are also more variable among uncovered workers, with $\sigma(w_{it}^T)$ about 12% higher than for covered workers.

An examination of coverage patterns within firms reveals that nearly all firms either have no covered workers or 100% union coverage. Firms with no coverage tend to be larger than covered firms (mean employment is 8.4 workers versus 6.1 for covered firms), and have substantially higher annual sales per worker (~74,800 versus ~39,900 for covered firms). They are also more likely to be located in Lisbon and to be in the non-financial services sector (42% versus 20% for uncovered firms).

The positive wage advantage for *uncovered* workers in Portugal stands in sharp contrast to the patterns in the U.S., the U.K., and Canada, where union coverage is positively correlated with wages. For example, data presented by Card, Lemieux and Riddell (2004) show that the difference in mean log hourly wages between workers who are covered by collective agreements and those who are not is between 15% and 30% in all three countries (and between 5% and 25% controlling for gender, education, and experience).

4. Proximate Analysis of the Components of Wages

We now turn to a descriptive analysis of the role of the wage floors, wage cushions, and supplementary payments in determining wage differentials between groups and overall wage inequality. For this analysis (and all analysis in the remainder of the paper) we focus on the sample of person-year observations in QP with assigned wage floors, described in columns 7-9 of Table 1.

Figure 2 shows the distributions of relative wage floors and wage cushions by gender, pooling across all years of our sample. Panel a shows that many wage floors (especially for female workers) are within 5 percentage points of the minimum wage, though there is a long upper tail of floors. Therefore, throughout our analysis we will explore the connections between wage floors and the minimum wage, specifically when analyzing the determinants of negotiated wage floors, the effect of wage floors on actual wages, and decomposing the changes in real wages, in Sections 5.2, 6, and 8, respectively. A similar pattern holds for the distribution of wage cushions in panel b. Approximately 27% of males and 39% of females earn a base wage that is within 5 percentage points of their respective floor.¹⁸

18. Note that legally all wage floors are required to comply with the national minimum wage, apart from those for apprentices and handicapped workers who are excluded from our sample. Actual base wages should also comply with the respective wage floors, though in our data set there can be negative wage cushions, arising if the base wage is misreported to QP, or in certain cases like an extended sick leave. In

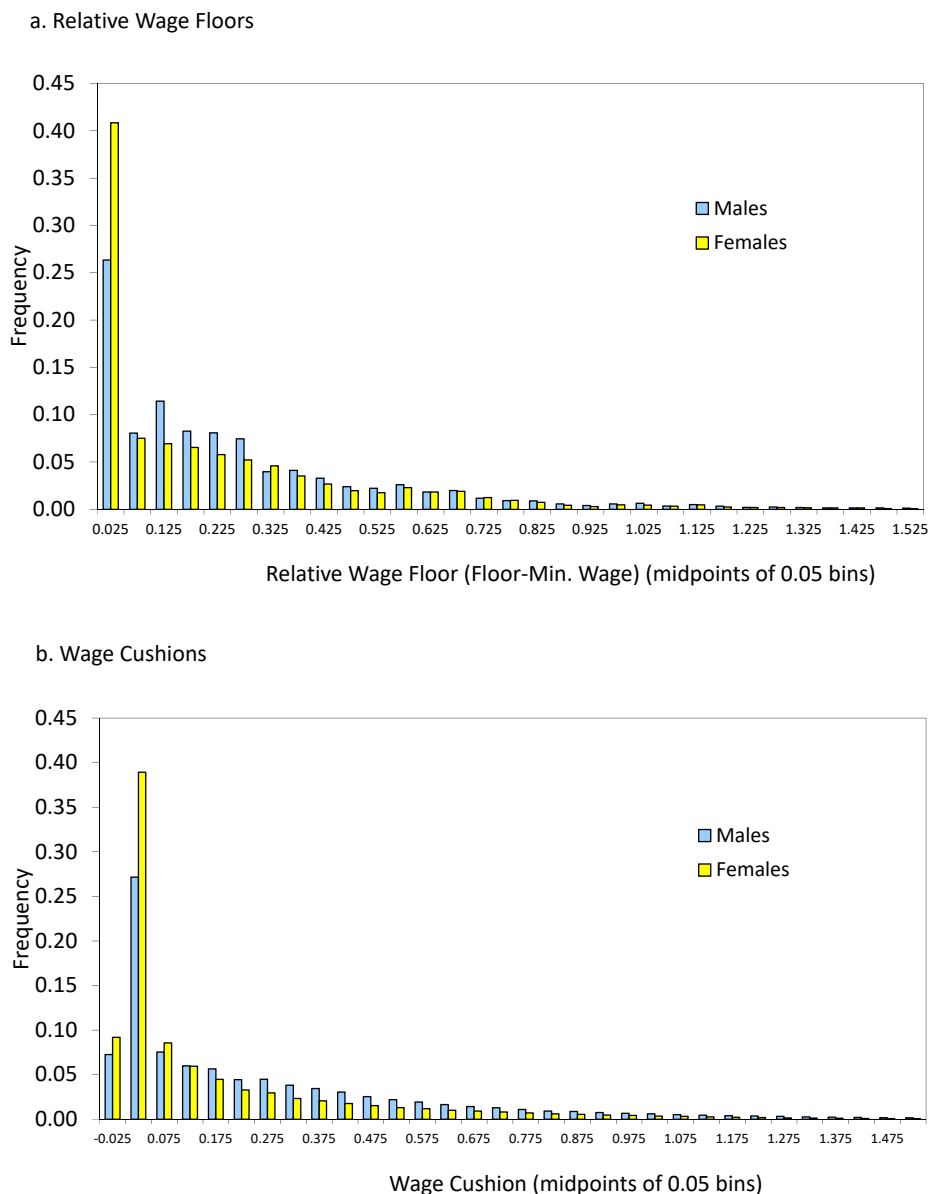


FIGURE 2. Distributions of relative wage floors and wage cushions by gender.

Figure 3 shows how the mean values of the three individual-specific components of wages highlighted in equation (2) vary over time for different groups of workers.

our analysis below we keep the small fraction of workers with negative wage cushions, except as noted in Section 6.

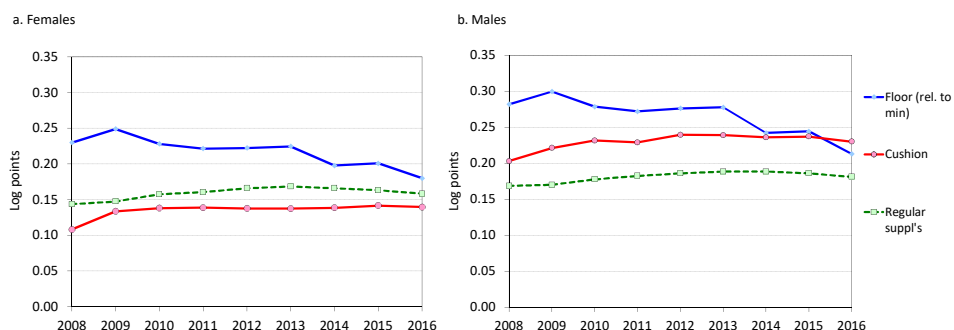


FIGURE 3.a. Components of mean wages (relative to minimum wage) for females and males.

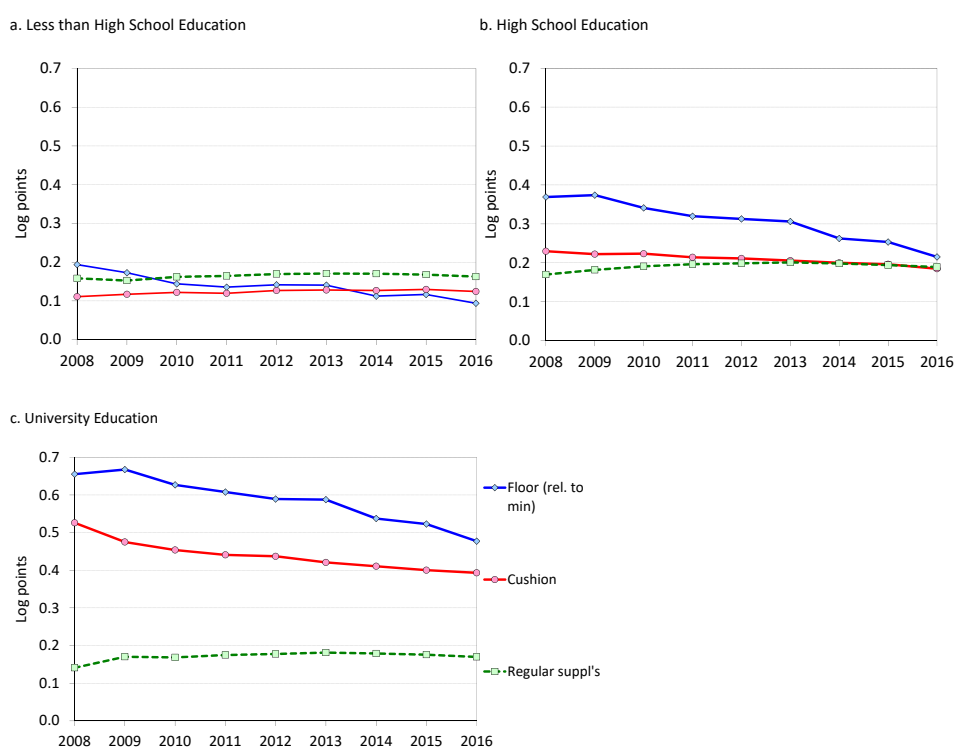


FIGURE 3.b. Components of mean wages (relative to minimum wage) by education group.

Figure 3a presents mean relative floors, cushions, and supplements by gender; Figure 3b present similar data by education level; and Figure 3c presents mean floors and cushions for workers employed at firms in different quartiles of the distribution of value added per worker. Finally, Figure 3d shows the mean values of wage floors, cushions, and supplements by age for female and male workers (pooling across all years in our sample).

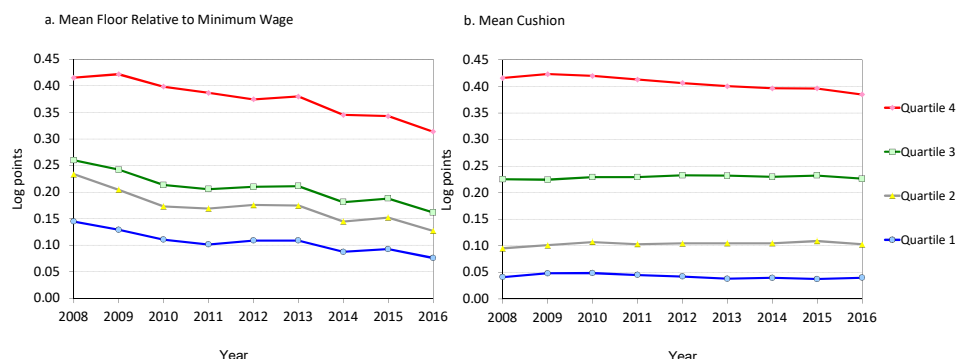


FIGURE 3.c. Mean floor (relative to minimum wage) and mean cushion by quartile of firm value added/worker.

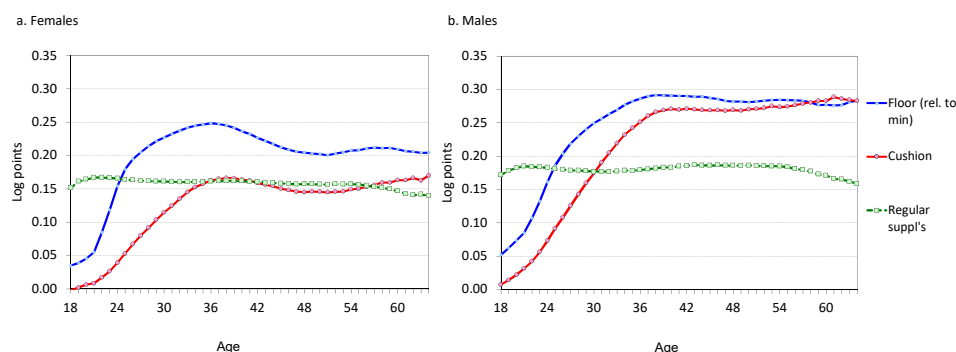


FIGURE 3.d. Components of age profile of mean wages (relative to minimum wage) for females and males.

In interpreting these figures it is helpful to keep two points in mind. First, our samples in 2008 and 2009 are slightly less representative than in later years, due to the lack of data on wage floors that were renegotiated one or two years ago prior to 2010. Second, the real value of the minimum wage rose relatively sharply between 2008 and 2010 (with a +6% adjustment in 2009 and a +4% adjustment in 2010). Thereafter the real minimum drifted downward for three years before raises of +4% in 2014 and +4% in 2016. Consequently between 2010 and 2016 the real minimum was relatively stable, ending up only 3 log points higher in 2016 than in 2010. Given these two factors we focus most of our attention on changes from 2010 to 2016 throughout this paper.

Examination of Figures 3a-3c shows that across most subgroups, mean relative wage floors fell during our sample period. The declines were partly driven by increases in the minimum wage, particularly in 2014 and 2016 (see Figure 3a). The declines were much larger for better-educated groups, but even for those with less than a high school education the mean gap between their wage floor and the minimum wage fell from about 15 log points in 2010 to 10 log points in 2016. As we show in Section 8 (below) a lot of the decline in relative wage floors for highly educated workers arose through a

re-allocation of workers to jobs with lower wage floors. Such re-allocations were less important for lower-educated groups, in part because their floors were clustered closer to the minimum wage even in 2010.

In contrast to the erosion in relative floors, the mean values of cushions and supplements were more stable, though there was a clear decline in mean cushions for workers with a university-level education. Figure 3c also shows that mean floors and cushions declined for workers at firms in the top quartile of value added per worker, leading to some narrowing of between-firm wage differentials —the opposite of the pattern documented for Germany (Card et al., 2013) and the U.S. (Song et al., 2019).

The age profiles in Figure 3d reveal that young workers tend to be employed in jobs with very low wage floors, and to receive small wage cushions. By age 25 or so, however, mean floors are in the 15% range and mean wage cushions are 5%-8%, and by age 40 a typical female has a wage floor of around 25 log points and a cushion of 15 points, while a typical male has a wage floor of nearly 30 points and a cushion of over 25 points. Thus, both floors and cushions contribute to the life cycle profile of wages.

Table 3 presents a more systematic summary of the net contributions of relative wage floors, cushions, and supplements to the levels and variances of wages, for all workers and by gender, education, and firm value-added quartile. The first 5 columns decompose the means of log salary, while columns 6-10 pertain to variances. In the first row, for example, we show the mean log real monthly wage for all workers, the mean wage differential relative to the minimum wage (61 log points), and the mean contribution of relative wage floors (24 log points), wage cushions (19 log points) and supplements (17 log points). As shown by the numbers in parentheses just below the row entries, these three terms contribute 40.2%, 31.4% and 28.4%, respectively, to the mean log gap between monthly salaries and the minimum wage.

For the decomposition of variances we show $\text{var}[rf_{it}]$, $\text{var}[h_{it}]$, $\text{var}[s_{it}]$, and $2\text{cov}[rf_{it}, h_{it}]$ (2 times the covariance of relative floors and wage cushions) which is nearly all cases the largest of the covariance terms arising from a decomposition of $\text{var}[w_{it}]$ based on equation (2). For the workforce as a whole, relative floors contribute 32.3% of the overall variance in log total salaries, cushions contribute 42.1%, supplements contribute 9.1%, and the positive covariance of floors and cushions across workers contributes 15.6%. Together these 4 terms account for 99.2% of the total variance.

The next set of rows in Table 3 show similar statistics for males, females, and for the gender gaps in mean log wages and the variance of wages. Males have higher and more variable wages than females, differences that are attributable to the both higher and more variable floors (30% of the gender gap in mean wages, 22% of the gap in variance of wages) and to higher and more variable cushions (60% of the gender gap in mean wages, 75% of the gap in variance of wages). Similar conclusions apply to the wages of more versus less educated workers. For example, 58% of the 27 log point gap in mean wages between high school graduates and those with less than a high school education is attributable to higher floors, while 31% is attributable to higher cushions. In their analysis of base wages, floors and cushions in the 1999 QP, Cardoso

and Portugal (2005) likewise found that cushions contributed about 40% of the overall return to each additional year of education.

As shown in the bottom three rows of the table, differences in mean wage cushions also play a significant role in explaining the mean wage gap between firms in the top and bottom quartile of value added per worker: just over one-half of the 68 log point gap in mean wages is explained by higher average cushions at more productive firms. Again, this is consistent with findings by Cardoso and Portugal (2005) on the effects of firm-specific productivity on the floor and cushion components of wages. These patterns suggest that the wage cushions at a firm are correlated with the firms' ability to pay, consistent with models of firm-specific wage setting (e.g., Card et al., 2018).

To investigate this more fully we conducted a simple analysis of wage floors, wage cushions, and log base wages, controlling for the specific CBA covering each worker. Specifically we fit models of the form

$$y_{it} = \alpha_0 + \alpha_x X_{it} + \sum_{d=2}^{10} \alpha_d I_{d(i,t)} + \psi_{CBA(i,t)} + u_{it} \quad (3)$$

where y_{it} represents either the wage floor, wage cushion, or log base wage of worker i in year t , X_{it} is a set of worker characteristics (gender, education and age), $d(i, t)$ is an index function that maps the worker to the value-added deciles of his/her employer in year t , I_d is a dummy for the d^{th} decile of value added, $CBA(i, t)$ is another index function that maps the worker to the specific collective bargaining agreement he or she is covered by in year t , and ψ_C represent a set of CBA fixed effects. The results are presented in Table 4. For reference we show models for the 3 outcomes with and without CBA effects. A comparison of the estimated α_x and α_d coefficients between these models allows us to assess how much of the overall variation in each outcome across workers and firms is preserved within collective agreements.

The wage floor models in columns 1 and 2 of Table 4 summarize the *assignment process* between workers and floors in the labor market as whole (column 1) and within CBA's (column 2). The coefficients of the worker-specific characteristics suggest that women are assigned to lower floors, while better-educated and older workers are assigned to higher floors: these matching effects are only somewhat attenuated within CBA's. The value added decile effects show that wage floors are higher at firms with higher value added per worker. Unlike the pattern for worker characteristics, however, the cross-firm gradient in floors is substantially flatter within CBA's than in the market as a whole. As emphasized by Boeri et al. (2021), this could be a cause for concern if less profitable firms are covered by collective agreements with relatively high floors.

The wage cushion models in columns 3 and 4 suggest that mean cushions vary across gender, education and age groups more or less the same within CBA's as they do in the labor market as a whole. In contrast, mean cushions are *more responsive* to firm profitability within CBA's, partly undoing the relatively flattening of differences in wage floors within agreements.

As a consequence, looking at the models for log base wages in columns 5 and 6 we see that about 85% of the market-level variation in mean base wages across value-added deciles is preserved *within CBA's*.¹⁹ In other words, sectoral agreements appear to only modestly dampen the sensitivity of wages to firm profitability.

The importance of cushions as a source of flexibility is summarized in columns 7 and 8. Since the base wage is just the sum of the wage floor and the wage cushion, we can calculate the share of the α coefficients reported in columns 5 and 6 that is attributable to the variation in cushions. This is around 60% for gender, education and age, regardless of whether we condition on CBA effects or not. It is closer to 70% for the coefficients associated with firm profitability deciles when we do not condition on CBA effects, but rises to around 90% when we look at wages within CBA's. Descriptively, then, wage cushions play a relatively large role in maintaining wage flexibility in the presence of sectorally bargained wage floors.

5. Determinants of Negotiated Wage Floors

In this section we turn to an analysis of the determinants of negotiated wage floor increases. Our first goal is to try to understand how the wage floors within a given CBA move relative to each other. To foreshadow our results, we find that in nearly all cases the floors are adjusted proportionally, so there is a single number—representing the mean increment in wage floors—that fully summarizes the negotiation results. Our second goal is to study how the rate of adjustment of wage floors responds to demand conditions at the firms covered by the CBA. We note that all our models include year effects, so we are not able to address the question of how collectively bargained wage floors adjust to changes in the minimum wage (as in Fougère et al., 2018).

5.1. Simple Models of Wage Floor Adjustment

As a starting point, consider a series of increasingly rich models for the change in the real wage floor of group g when CBA c is renegotiated in year t :

$$\Delta f_{cgt} = \delta_t + \varepsilon_{cgt} \quad (4a)$$

$$= \delta_t + Z_{ct}\delta_Z + \varepsilon_{cgt} \quad (4b)$$

$$= \delta_{ct} + \varepsilon_{cgt} \quad (4c)$$

$$= \delta_{ct} + R_{cgt}\delta_R + \varepsilon_{cgt} \quad (4d)$$

(Note that Δf_{cgt} involves a change over different numbers of years, depending on when contract c was last negotiated). Model (4a) includes only year effects: the fit of such a model allows us to assess how far are CBA renegotiations in Portugal from the “fully centralized” benchmark that is often taken as a normative ideal by

19. This estimate comes from regressing the estimated decile effects in column 6 on those in column 5.

macroeconomists (e.g., Calmfors and Driffill, 1988). Model (4b) adds some contract-specific characteristics Z_{ct} —most importantly, the duration of time since the last negotiation, which can range from 1 year to 3 years, or even longer in a few cases. Model (4c) includes a set of contract-specific fixed effects, which fully absorb *any* CBA $\times$ year factors, like industry-wide demand shocks or changes in local labor market conditions that affect workers in the contract. The fit of this model allows us to assess the extent to which all floors within a given CBA move together. Finally, model (4d) adds a set of characteristics R_{cgt} of the workers covered by wage floor c, g , and asks whether there is any evidence that floors within the same CBA are adjusted to reflect the characteristics of the covered workers in different floor groups, controlling for the mean contract-level change in floors (captured by δ_{ct}).

Table 5 presents adjusted R^2 statistics for variants of these 4 sets of models, estimated using the changes in mean real wage floors for workers in our matched QP-BTE data base.²⁰ We estimate the models by weighted OLS, using as weights the number of workers in floor group g of CBA c .

The fit of fully centralized model (4a) (row 1) is surprisingly good, with an adjusted R^2 statistic of close to 80%, suggesting that most of the variation in wage floor adjustments is explained by just 7 year effects. Adding controls for industry, worker characteristics and time since last negotiation (row 4) raises the adjusted R^2 to 85%; adding industry $\times$ year effects (row 5) raises it to nearly 90%.

The specification in row 6 adds contract-year effects (i.e., model 4c). These increase the adjusted R^2 to 98%, leaving almost no unexplained within-contract variation in wage floors. Adding controls for the mean fraction female, mean age and mean education of workers covered by each floor (model 4d) increases the fit only very slightly. These variables have very small but statistically significant effects, showing slightly faster growth in floors that cover a higher share of women, older workers, and less-educated workers.

Overall we reach three conclusions from this simple analysis. First, nearly 80% of the variation in average negotiated wage floors across our sample is explained by year effects. Second, about one half of remaining variation is explained by industry-specific shocks, workforce demographics, and the lag since the last negotiation. Third, nearly all the remaining variation is explained by CBA-specific effects, meaning that in a typical negotiation all the floors are adjusted by the same percentage.

5.2. Modeling Contract-Wide Mean Changes in Floors

Building on the findings in Table 5, we turn to an analysis of the determinants of the mean wage floor adjustment, δ_{ct} in a given sectoral contract negotiation (estimated from a model like equation 4c). Our main focus is on the question of how wage floor increases are affected by changes in productivity/profitability of firms covered by the

20. Note that we exclude wage floors that are set at exactly the minimum wage, since adjustments for such floors are presumably insensitive to firm, industry, or worker characteristics.

CBA. In particular, we are interested in whether firms with faster productivity growth exert a stronger influence on negotiations than less profitable firms (Boeri et al., 2021; Fanfani, 2020), potentially threatening the survival of the latter firms. We proceed by examining the effects of changes in the mean and various quantiles of the distribution of value added per worker among the firms affected by the contract.

Let $m(VA_{ct})$ represent the mean of log value added per worker in year t for firms covered by sectoral CBA c , and let $q(VA_{ct})$ represent the employment-weighted q^{th} quantile (e.g., the 25th or 50th) of the distribution of log value added among these firms. Assume that contract c is renegotiated in year t and was last renegotiated in $t - \ell$ (so $\ell = 1, 2, 3$ is the years since last renegotiation). Then the changes in productivity relevant for the renegotiation can be summarized by:

$$\begin{aligned} DmVA_{ct} &= m(VA_{jct-1}) - m(VA_{jct-\ell-1}) \\ DqVA_{ct} &= q(VA_{jct-1}) - q(VA_{jct-\ell-1}). \end{aligned}$$

Note that we lag the financial information by a year, reflecting the fact that a contract that is updated in year t will have been negotiated before financial information from the current year is realized.²¹

Our first set of models for floor adjustments in contract c , presented in Table 6, take the form:

$$\delta_{ct} = \beta_0 + \beta_1 DxVA_{ct} + \beta_2 Z_{ct} + e_{ct} \quad (5)$$

where $DxVA_{ct}$ is the change in the mean ($x = m$) or some quantile ($x = q$) of the distribution of value added per worker among relevant firms, and Z_{ct} are a set of contract-specific covariates, including time effects, dummies for the number of years since the last renegotiation, a measure of cumulative inflation since the last negotiation, and measures of the share of females, the share of university graduates, and the mean age of workers covered by the contract. Column 1 shows a model using $DmeanVA_{ct}$ as the measure of demand-side factors, while columns 2-6 replace this with $DqVA_{ct}$ based on the 10th, 25th, 50th, 75th, and 90th quantiles.

Judging by the adjusted R-squared statistic, the change in the *mean* of value added among firms covered by a CBA is the best predictor of negotiated wage floor changes, though the median is a close second. The magnitude of the estimated β_1 coefficient suggests that wage floors are relatively responsive to the central tendency in industry-wide productivity growth, with an elasticity of wages to mean or median changes in value added of around 0.06-0.07. There also appears to be some limited “catch up” for past inflation: the model in column 1, for example, implies that real wages recover about one-fifth of their lost value arising from inflation since the last negotiation.²²

21. Note that we use the change in the q^{th} quantile of VA_{jct} , rather than the q^{th} quantile of the change in VA_{jct} , to summarize the distribution of demand shocks among firms covered by a given contract. Under the rank invariance assumption that is widely used in the quantile treatment effects literature (e.g., Firpo 2007) these are the same.

22. Since all the models in Table 6 include year effects and dummies for the number of years since the last negotiation, the identification of the lagged inflation effect relies on differences in inflation over different time windows in our sample period, similar to Card (1990).

Finally, consistent with the evidence in Figures 3a and 3b, floors covering a higher share of female workers tended to rise more quickly in the 2010–2016 period, while floors covering a higher share of university-educated workers tended to fall slightly.

A potential concern with the models in Table 6 is that changes in firm-specific value added contain transitory fluctuations (and/or measurement errors) that are not completely eliminated by using the industry-wide means or medians. Such fluctuations/errors may play an outsize role in driving measured changes in the upper and lower quantiles. To address this issue, we conducted a complementary analysis of longer-run changes in contractual wage floors. Specifically, for all CBA's that were renegotiated at least once between 2010 and 2016, we constructed the average change in wage floors from 2010 to 2016, then fit a series of models relating this longer-run change to corresponding changes in the mean and quantiles of value added per worker for firms covered by the CBA. The results are presented in Table 7. We present specifications with no other controls in the upper panel, and models that control for the modal industry of the covered firms (with a total of 7 dummies) in the lower panel.

As expected, these models show a somewhat higher elasticity of wage floors with respect to productivity changes among covered firms, with a point estimate of 0.134 for the effect of the change in mean log value added when major industry dummies are excluded from the model and 0.093 when they are included. As in Table 6, the best fitting models are those that relate changes in wage floors to changes in the central tendency of productivity change among covered firms. Moreover, if we estimate models that include *both* the median (or mean) change in value added and one of the other quantiles (see Online Appendix Table D.1), we find that all the explanatory power comes from the median or mean change.

The magnitudes of the estimated elasticity of wage floors with respect to mean or median changes in value-added in Table 7 are comparable or larger than typical estimates in the rent sharing literature (see Card et al., 2018). We note, however, that if changes in wage floors lead to some compression of wage cushions (as we find to be the case in the next section) then the impact of value added changes on average *wages* will be smaller than the impact on wage floors. In fact, we find that only about one-half of a rise in wage floors is passed on to wages —the other half is absorbed by reductions in wage cushions. Assuming a 50% passthrough the implied elasticity of workers' wages with respect to rises in productivity among firms covered by the relevant CBA is between 4.6% and 6.7% —closer to the middle of the range of estimates in the rent-sharing literature.

6. Effect of Wage Floors on Wages

In this section we turn to an analysis of the effect of changes in wage floors on the actual wages of workers. Conceptually, our approach builds on standard techniques for studying the passthrough of an increase in sales taxes to the final price paid by consumers. Specifically, we relate actual wage changes for workers at a firm to the

simulated changes that would occur if floors were adjusted but all other components of wages remained fixed.

Consider the set of employees at a given firm j in year $t - 1$. Let Δf_{it} represent the percentage change in the real wage floor between $t - 1$ and t for worker i in this set. If the CBA covering the worker was renegotiated in the past year then Δf_{it} is just the negotiated floor adjustment in that contract (adjusted for inflation). If the CBA was not renegotiated then Δf_{it} is minus the percentage change in the price deflator between $t - 1$ and t . Using the notation introduced in Section 3, let W_{it-1} represent the level of the monthly base wage of the worker in year $t - 1$ and let F_{it-1} represent the level of her wage floor. We define:

$$\begin{aligned}\Delta w_{it}^* &\equiv \ln(W_{it-1} + F_{it-1}\Delta f_{it}) - \ln(W_{it-1}) \\ &\approx (F_{it-1}/W_{it-1})\Delta f_{it},\end{aligned}\tag{6}$$

which is just the simulated increase in the log base wage of worker i if her wage floor were increased by the proportion Δf_{it} and there was no change in the gap between her base wage and her wage floor.

The actual change in the worker's base wage includes the change in H_{it} , the gap (in Euros) between her base wage and her floor:

$$\begin{aligned}\Delta w_{it} &= \ln(W_{it-1} + F_{it-1}\Delta f_{it} + \Delta H_{it}) - \ln(W_{it-1}) \\ &\approx (F_{it-1}/W_{it-1})(\Delta f_{it} + \Delta H_{it}/F_{it-1}) \\ &= \Delta w_{it}^*(1 + \gamma_{it})\end{aligned}\tag{7}$$

where $\gamma_{it} = \Delta H_{it}/\Delta F_{it}$ is the ratio of the change in the absolute cushion component for worker i to the absolute change in her wage floor. To illustrate the implications of this equation, consider two limiting cases. At one extreme, suppose that H_{it} remains constant as the wage floor changes (as is assumed in the construction of Δw_{it}^*). Under this scenario $\gamma_{it} = 0$, and (7) implies that $\Delta w_{it} = \Delta w_{it}^*$. At the opposite extreme, suppose that the base wage W_{it} remains constant as the wage floor is raised (a situation that can only happen if $W_{it-1} > F_{it}$ —i.e., the initial base wage is above the new floor). Under this scenario, $\gamma_{it} = -1$ and $\Delta w_{it} = 0$ —i.e., an increase in floors is fully offset by a reduction in the worker's wage cushion.²³ Note that we can also construct a parallel measure of the effect of floor increases on a worker's *total wage* under the assumption that the gap between the worker's total wage and her floor stays constant, and compare that to the change in her total wage, Δw_{it}^T .

To proceed, consider a simple regression model relating Δw_{it} to Δw_{it}^* and a set of controls (X_{it}):

$$\Delta w_{it} = \theta_0 + \theta_1 \Delta w_{it}^* + \theta_x X_{it} + \xi_{it}\tag{8}$$

We focus on estimating this model for the set of workers who remain at the firm between $t - 1$ and t and stay in the same wage floor group—a group we refer to as

23. A third scenario is one in which each worker's *proportional* cushion h_{it} remains constant as floors change. In this case $\Delta w_{it} = \Delta f_{it}$.

the “firm stayers”. The coefficient θ_1 provides a measure of the effect of wage floors on the base wage of stayers. A salient null hypothesis is $\theta_1 = 1$, which corresponds to the hypothesis that increases in wage floors are passed through fully to workers. If rising wage floors are partially absorbed by a reduction in wage cushions, however, then $\theta_1 < 1$, and in the limiting case in which floor increases have no effect on wages, $\theta_1 = 0$.

A potential concern in estimating a model of wage changes for stayers is that workers who remain with the firm and in the same floor category may be selected in a way that is correlated with their potential wage increase, leading to selection bias in the error term ξ_{it} . To address this, and to set the stage for the employment growth models we present in the next section, we present instrumental variable (IV) estimates that use $\Delta \bar{w}_{jt}^*$ (the mean of Δw_{it}^* across all N_{jt-1} employees of the worker’s firm in year $t - 1$, including stayers and non-stayers) as an instrument for the worker-specific simulated wage increase.

Estimation results for a variety of specifications of equation (8) are presented in Table 8. Columns 1-4 present models for the effect of floors on base wages while columns 5-8 present a parallel set of models for total wages. As a point of departure, columns 1 and 5 present simple OLS models based on equation (8). The control variables include year effects and dummies for female gender and university education, as well as a linear term in the worker’s age. We also add the change in log real value-added per worker at the employer. This is meant to control for firm-specific demand shocks that may be jointly correlated with the unexplained component of base wage increases (i.e., ξ_{it}) and the increase in wage floors affecting the firm.

The models for base wages (column 1) and total wages (column 5) yield estimates of $\theta_1 \approx 0.45$; in both cases the estimates are relatively precise. Corresponding IV models that use $\Delta \bar{w}_{jt}^*$ as an instrumental variable for Δw_{it}^* are presented in columns 2 and 6. The estimated first stage effects of $\Delta \bar{w}_{jt}^*$ on Δw_{it}^* are reported in the second last row of the table: in both cases the first stage coefficients are close to 1.0 in magnitude and highly significant. Interestingly, the IV estimates of θ_1 are about 15% larger in magnitude than the OLS estimates, suggesting that ξ_{it} is negatively correlated with Δw_{it}^* (perhaps reflecting the omission from the sample of workers who get promoted to a higher floor group).

Given that the first stage coefficient of $\Delta \bar{w}_{jt}^*$ is close to 1, the IV estimates of θ_1 in columns 2 and 5 are (approximately) equal to the reduced form effects of $\Delta \bar{w}_{jt}^*$ on Δw_{it} or Δw_{it}^T . Moreover, in the absence of individual-level covariates these reduced-form effects would be numerically equivalent to the effects obtained from a *firm-level* regression model relating the average wage increase for all stayers at the firm ($\Delta \bar{w}_{jt}$) to $\Delta \bar{w}_{jt}^*$ and controls:

$$\Delta \bar{w}_{jt} = \rho_0 + \rho_1 \Delta \bar{w}_{jt}^* + \rho_x X_{jt} + \bar{\xi}_{jt} \quad (9)$$

Our individual-level models include individual-specific gender, education and age controls so we cannot quite reproduce the micro-level estimates from the firm level regression. However, as shown in columns 3 and 7, when we estimate equation (9)

using firm-wide averages of the covariates as controls we find, as expected, that the estimates of ρ_1 are approximately equal to the corresponding IV estimates of θ_1 .²⁴

Finally, the specifications in columns 4 and 8 interact $\Delta \bar{w}_{jt}^*$ with a variable indicating the fraction of all workers at the firm whose wage floor was renegotiated between $t - 1$ and t . (For the 90% of firms in which all workers are covered by a single CBA, this fraction is either 0 or 1, depending on whether the CBA was recently renegotiated or not, but for firms where different occupation groups are covered by different CBA's it can be strictly between 0 and 1). This interaction term allows us to check whether the responsiveness of wages to floor changes is the same when wage floors are explicitly adjusted upward by a contract renegotiation as when they are passively adjusted (typically downward) by inflation.²⁵ The estimated interaction effects are statistically indistinguishable from 0, providing no evidence of asymmetry in the passthrough of wage floor changes.

The estimates in Table 8 suggest that on average only about one-half of the implied increases in wages arising from changes in wage floors are passed through to workers. The balance is offset by reductions in wage cushions, with relatively small effects on supplementary wage payments, given the similarity of the passthrough effects on base wages and total wages. The ability of any particular worker's wage cushion to absorb an increase in wage floors, however, depends on the size of their wage cushion. In the minimum wage literature, for example, a typical finding is increases in minimum wages have large effects on workers whose wage is below or close to the new minimum but smaller or even zero effects on those earning substantially above the minimum.²⁶

We explore the heterogeneity in the degree of passthrough of wage floor increases in Table 9. Each row represents a different skill group (classified by gender, education, and age). For each group we report the share of all firm stayers in the group, the mean relative wage floor, mean wage cushion and mean supplements for members of the group (as measured in year $t - 1$), and group-specific estimates of the passthrough effect based on the aggregated reduced form model of equation (9). We exclude results for people age 18-24 with a university education because for both genders this group is extremely small.

As expected, the estimated passthrough rates tend to be larger for groups with lower wage cushions. For example, females with less than high school education who are between 25 and 44 have an average wage cushion of roughly 8 log points, and estimated passthrough rates of 0.71 (st. err=0.02) using base wages or 0.63 (st. err=0.07) using total monthly wages. By comparison, females in the same age range with a university education have an average wage cushion of roughly 48 log points, and estimated passthrough rates of 0.31 (st. err=0.05) and 0.37 (st. err=0.08).

24. The standard errors are about the same too, which is expected given that we cluster the standard errors by firm.

25. During our sample period there were two years with negative inflation in Portugal.

26. See Fortin, Lemieux, and Lloyd (2021) and Cengiz et al. (2019) for overviews of the existing literature and evidence on spillover effects of minimum wages in the U.S. labor market.

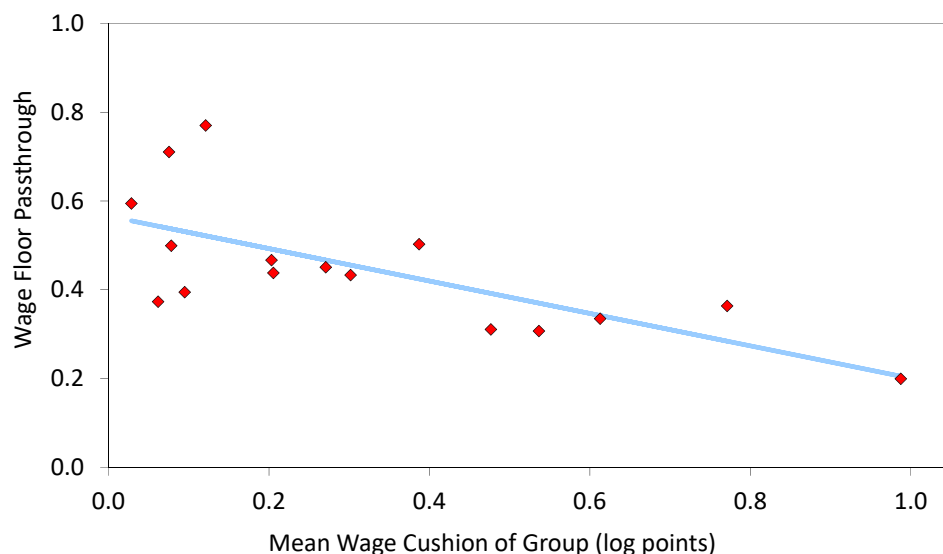


FIGURE 4. Estimated passthrough rates of floor increases to base wages, by group. Based on estimates in Table 8. Fitted OLS line shown, R -squared = 0.49.

This relationship is illustrated in Figure 4, where we plot the estimated passthrough rate for each demographic group (estimated using base wages) against the mean wage cushion for the group. We draw two conclusions from this graph. First, even for low-cushion groups the passthrough rate is less than 1, suggesting that the modest floor increases typically negotiated during our sample period (in the range of 1%-3%) were partly absorbed by compressing wage cushions. Second, in contrast to the pattern found in the minimum wage literature, wage floor increases in collective bargaining agreements appear to have some positive “spillover” effect even on relatively high-cushion groups.

Robustness Checks. The passthrough models in Table 8 address concerns about the selectivity of workers who remain in the same firm and floor category. Another potential concern is the endogeneity of the wage floor itself. Unobserved demand shocks might influence both the evolution of wage floors and a firm’s willingness to raise wages. To investigate this issue we use an IV procedure that links back to Section 5.2 on the determinants of floor changes. Specifically, in the first stage, we instrument the change in wage floors (for renegotiated CBA’s) using the average change in value added per worker at other firms covered by the CBA since the time of the last agreement. Given the lags in contract renegotiations, we impose the requirement that lagged information on value added per worker is available for at least 2 previous years for the firm and other firms in the CBA.

Results are reported in Online Appendix Table E.1. Importantly, we find that estimates of wage floor passthrough from specifications similar to those in Table 8 are very similar in the restricted sample with lagged value added data. When we

use changes in value added per worker at other firms as an instrument for the wage floor increases, we obtain larger estimates of the passthrough effect than from our base specifications.²⁷ We emphasize, however, that value added changes at other firms in the same CBA may be correlated with sectoral demand shocks that positively effect wages, leading to an upward bias in this alternative IV strategy.²⁸

Next, we strengthen the link of our analysis to the minimum wage literature. We complement our analysis of the impacts of wage floors on different demographic subgroups by dividing workers into different groups based on the size of their wage cushions prior to the contract renegotiation. This approach is very similar to that used in Dustmann et al. (2022) to study the spillover effects of the newly introduced German minimum wage. Online Appendix Figure E.1 shows the effect of a rise in the wage floor on wages for workers whose base wage was closer or further from the old floor. Consistent with the patterns in Figure 4, we find larger passthrough effects for workers whose base wage was close to the floor (i.e., that had a low ‘wage cushion’). But in contrast to the minimum wage literature, we find that the “spillover effect” of base wage increases extends to workers earning considerably above the old wage floor – again consistent with the patterns in Figure 4.

7. Effect of Wage Floors on Employment

Although increases in wage floors are partly absorbed by the compression of wage cushions, they still lead to some increase in workers’ base wages and total salaries. This opens up the question of whether firms use reductions in employment as another channel of adjustment to higher floors. Such employment effects might be expected if employment and wage outcomes lie on a traditional downward-sloping employment demand function. To the extent that wages are endogenously set by firms with market power, however, the equilibrium relationship is less clear and may even be upward-sloping.

Building on the results in Table 8 we fit a series of simple models of the form:

$$\Delta \ln E_{jt} = \tau_0 + \tau_1 \Delta \bar{w}_{jt}^* + \tau_x X_{jt} + \zeta_{jt} \quad (10)$$

where E_{jt} is the total number of employees of firm j in year t and $\Delta \bar{w}_{jt}^*$ is the average simulated change in total base wages of employees present at the firm in period $t - 1$. We estimate this model for all firms, and separately for the subset where the modal

27. In these models we follow the specifications in Section 5.2 and include dummies for time since the last contract renegotiation and a measure of cumulative inflation over that period.

28. This is particularly a concern because firm-specific value added measures are noisy, and value-added changes at other firms may proxy for firm-specific changes.

worker is covered by a *sectoral* CBA.²⁹ For the latter set of firms wage floors are arguably more exogenous to firm-specific conditions.

The results are presented in Table 10. We show a specification with only year effects and a control for the increase in real value added per worker at the firm in columns 1 and 4, a second model with controls for gender, fraction of university educated workers, and mean age of workers in columns 2 and 5, and a specification that allows for an interaction between $\Delta \bar{w}_{jt}^*$ and the share of workers with a renegotiated wage floor in columns 3 and 6.

The estimation results are quite similar for the models estimated on all firms and on the subset covered by sectoral contracts, and point to three main conclusions. First, increases in firm-specific productivity (as measured by the change in real value added per worker) have a significant positive effect on employment growth, about ten times larger in magnitude than the effect on wages.

Second, none of the models show a negative effect of floor increases on employment growth. The model in column 2, for example, yields an estimate of $\hat{\tau}_1 = 0.165$ for the effect of floor-induced base wage increases on employment. If one assumes that firms set employment taking wages as exogenous then one could convert this into an estimated demand elasticity by dividing by the estimate of ρ_1 from equation (9). Using the estimate of ρ_1 from column 2 of Table 8 yields an estimate of the elasticity of employment with respect to base wages of 0.30, with a standard error of approximately 0.33. While this point estimate is positive, a 95% confidence interval ranges from -0.36 to 0.96, so we cannot rule out small negative employment responses.

A third finding, consistent with the results in Table 8, is that there is no evidence of asymmetry in reactions to actively renegotiated wage floor changes versus changes in real wage floors arising from inflation.

The models in Table 10 describe employment outcomes for all workers. The results in Table 9, however, suggest that the wage impacts of wage floor increases vary across groups. To check whether there is similar heterogeneity in the employment impacts, we estimated models like (10) by gender, education, and age group. The results are presented in Online Appendix Table E.2, alongside the corresponding estimates of the wage effects for each group from Table 9. Ten of the 16 estimated employment effects are positive while six are negative. Only one is significantly negative ($t=2.02$); three are significantly positive ($t=2.43, 2.97, 4.19$). Moreover, the estimated employment effects for each group are positively correlated with the corresponding wage passthrough effects—the opposite of what would be expected if wage floors have larger negative effects on the employment of groups whose wages are most responsive to floor increases.³⁰

Since young workers have small average wage cushions (see Figure 3d), the findings in Table 9 suggest that their wages may have been pushed up relatively more

29. As noted, about 90% of firms have only a single CBA, but for firms where workers are covered by 2 or more different CBA's we assign sectoral coverage status based on the characteristics of the agreement that covers the largest number of workers.

30. The correlation of $\hat{\tau}_1$ and $\hat{\rho}_1$ across the 16 groups is 0.21.

by increases in wage floors over the past decade, preventing firms from hiring them in the first place—an effect that may be hard to discern from models of employment growth such as equation (10). While a full analysis of this concern is beyond our scope, Online Appendix Figure E.2 shows data on the fractions of young men and young women (age 16-24) who were not in employment, education or training in Portugal and 6 other countries (Italy, U.S., Spain, France, U.K., and Germany) over the 2004-2019 period. The so-called “NEET” rates for both gender groups in Portugal track the rates in other countries fairly closely: there is not much evidence of a relative rise in the post-crisis era. For example, comparing Portugal to the U.S., the difference in differences of NEET rates for 2017-2019 versus 2004-2007 is -0.3% for males and +1.4% for females. Parallel differences of differences relative to the U.K. are -4.4% for young men and -3.7% for young women. The only country that did appreciably better than Portugal (and virtually all other countries) was Germany.

8. Decomposing Changes in Real Wages, 2010-2016

In this section we combine the insights from the previous sections and document how the various components of wages contributed to overall changes in wages for the economy as a whole and for different groups over the 2010-2016 period. Our approach builds on the methodology developed by DiNardo, Fortin and Lemieux (1996)—hereafter DFL—for analyzing the effects of trade unions and minimum wages on trends in U.S. wage inequality. Specifically, we conduct a series of counterfactual simulations—summarized in Table 11—that provide a step-by-step decomposition of the changes in mean total monthly real wages for different groups of workers.

We start with scenario A, which takes all workers in our matched QP-BTE sample in 2010. Outcomes in this sample represent the actual distribution of wages in 2010. Next, in scenario B, we increment the wage floor that applies to each worker in 2010 by the percentage change of that floor between 2010 and 2016, holding constant the worker’s (proportional) wage cushion and (proportional) wage supplements. A comparison of outcomes between scenario B and the baseline scenario A allows us to assess what would have happened if floors adjusted as they did between 2010 and 2016, but all workers remained in their same floor categories, and received the same cushions and supplements as they did in 2010.

In scenario C, we reweight the observations in scenario B by the relative probability that workers in a given gender/education/age cell were present in the labor market in 2016 versus 2010. Following the logic of DFL, this reweighting allows us to assess how the changing demographic composition of the workforce would have affected wage outcomes, holding constant the assignment of workers to their 2010 wage floor groups, with their 2010 cushions and supplements, but with 2016 floors.

In scenario D, we take all workers in our matched QP-BTE sample in 2016, but assign each worker in a given wage floor group a randomly drawn wage cushion and

wage supplement from the distributions of *the same wage floor group in 2010*.³¹ Relative to scenario C (which has 2010 workers in their 2010 floor groups but assigned the 2016 floors) scenario D captures any reallocation of workers across wage floor groups, while holding constant wage floors at their 2016 values, and the distributions of wage cushions and supplements for workers in a given floor group at their 2010 distributions.

We note that this reallocation effect reflects a combination of within-job effects, between-job effects, and entry effects. Within jobs, a change in the rate at which workers are promoted to higher wage floor categories will lead workers in a given age range in 2016 to be assigned to better or worse wage floor groups than they would have been assigned to in 2010. For job changers, any shift in the probabilities of moving up or down the “job ladder” (as measured by the level of the wage floors at the origin and destination job) will likewise lead to a change in the assignment of workers to wage floors. Finally, any change in the assignment of labor market entrants (or re-entrants) to wage floor groups will contribute to the overall reallocation effect.

In scenario E, we adjust scenario D by assigning each worker his or her *actual* wage cushion in 2016. A comparison with scenario D allows us to assess the impact of changes in the distribution of wage cushions within a given wage floor group. Finally, Scenario F just takes the distribution of workers in 2016 with their 2016 floors, cushions and supplements. This differs from scenario E by the updating of the distribution of wage supplements from 2010 to 2016, allowing us to quantify the impact of changing wage supplements.

Table 12 summarizes the comparisons across these different scenarios for the overall population of workers and various subgroups. We begin by showing the mean log total monthly wage in 2010 (column 1) and the components of this total, as described by equation (2) (columns 2-4). Next we show the actual change in mean log wages between 2010 and 2016, which was -1.7% for workers as a whole, but ranged between -20% (for some university-educated groups) to +0.7% (for women with less than high school education who were between 25 and 44).

Column 6 shows the difference in mean log wages between scenario B and scenario A, and summarizes the impact of changing wage floors. On average real wage floors declined by about 2.2%, but the mean floors affecting young and less educated workers actually rose slightly, reflecting the influence of the minimum wage, which increased in real value by 3.5% between 2010 and 2016, pushing up some of the lowest wage floors in the economy.³²

Column 7 shows the effect of demographic changes captured by the difference between scenario C and scenario B. (Note that within any of the narrowly defined gender/education/age groups in the bottom panel of the table this difference is 0).

31. This re-assignment approach builds on DFL, who assessed the effect of a national changing minimum wage by assigning the lower tail of wages from one year to the distribution of wages in another year.

32. The decline in wage floors was more pronounced for university educated workers. If they had kept their job categories of 2010, their real wage floors would have declined by about 4%.

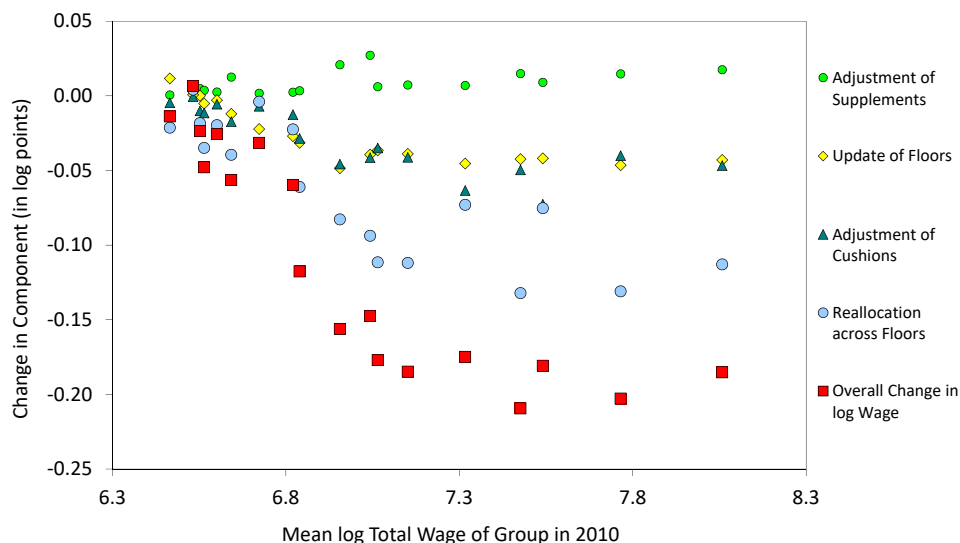


FIGURE 5. Components of change in mean log real wages across groups, 2010–2016.

Average education levels were rising quickly in Portugal between 2010 and 2016, a trend that would have increased wages by about 7.4% in the absence of other factors.

Offsetting the rise in education was a reallocation of workers across floor groups, the effects of which are captured by the differences in mean wages between scenarios D and C, presented in column 8. On average workers were being reallocated to lower-paying job over our sample period, leading to a nearly 5% reduction in real wages. The effects of this downgrading were particularly large for older university-educated and high-school educated workers, and were negligible for younger, less educated workers who were already working at jobs with the lowest wage floors. It is possible that this reallocation was partly caused by high wage floors, but our interpretation is that it was more likely a reflection of the fact that job openings at higher "rungs" of the job ladder tend to disappear in recessions (e.g., Moscarini and Postel-Vinay, 2018).

Next, columns 9 and 10 show the effects of changes in wage cushions and wage supplements. On average wage cushions declined over the course of the financial crisis, with larger declines for groups that were initially earning larger average cushions. In contrast, the value of wage supplements was relatively stable, though groups with the largest declines in floors and cushions experienced small increases in the value of their supplemental payments. This reflects the fact that some components of supplementary payments are expressed in absolute terms (such as meal allowance payments), and as the base wage of a group declines the *relative value* of their supplementary payments will rise.

The general pattern of the different components in columns 7–10 is illustrated in Figure 5. We plot the overall change in log wages for each of the 18 demographic groups highlighted in Table 12 against their mean log wage in 2010, along with the contributions of floor updates, changes in cushions and supplemental payments,

and the effect of re-allocations across floor groups. The figure shows that the large reductions in real wages for higher-paid groups in the Portugal between 2010 and 2016 reflected the combined effects of falling wage floors, reduced wage cushions, and re-allocations to lower-paying floor categories.

One way to summarize the relative contributions of these different components to the between-group pattern of wage changes is to compute $\text{cov}[\Delta \bar{w}_s^T, \Delta \bar{z}_{ks}] / \text{var}[\Delta \bar{w}_s^T]$, where $\Delta \bar{w}_s^T$ represents the change in the real average total wage of skill group s between 2010 and 2016, and $\Delta \bar{z}_{ks}$ is the mean change in the k th component for skill group s . Since $\Delta \bar{w}_s^T = \sum_k \Delta \bar{z}_{ks}$ these terms sum to 1. Following this approach we estimate that changes in real wage floors accounted for 24% of the between-skill group variation in real wages reductions, changes in real wage cushions accounted for 26%, re-allocation across floor group accounted for 56%, and changes in real supplemental payments accounted for -6%.

The final set of columns quantify the impact of the national minimum wage on wage floors. Column 11 shows the fraction of workers with a wage floor equal to the minimum wage in 2010. Column 12 reports the fraction whose floor would have equaled the minimum wage in 2016 if they had remained in the same floor group, subject only to floor updating. Finally, column 13 reports the fraction of workers with a wage floor equal to the minimum in 2016. By comparing columns 11 and 12 we can characterize the degree of “encroachment” of the minimum wage on lower wage floors. In 2010 13% of the workforce were in job categories with a wage floor exactly at the level of the minimum wage. Holding everyone in their 2010 jobs, this fraction would have risen to 19% by 2016. Thus, all else equal, about 6% of workers would have had wage floors that were overtaken by the minimum wage, leading to a roughly 50% increase in the share of workers with floors at the minimum. In addition, as shown by the comparison between columns 12 and 13, demographic changes and re-allocations of workers to job lead to another 1.5% increase in the fraction of workers with floors at the minimum wage. Looking across gender, age and education groups we see that the re-allocation effect is particularly large for younger workers in all education groups, reflecting the tendency for new entrants in all education categories to enter jobs with relatively low wage floors during the 2010-16 period. An interesting question for future research is whether this pattern was partly reversed in the recovery that has taken place since 2016.

9. Conclusions

In this paper we have tried to provide a simple framework for thinking about the effect of “European style” sectoral wage contracts on wage inequality and patterns of wage changes for different individuals and groups over time. Our approach builds on earlier work by Cardoso and Portugal (2005). As they (and many subsequent authors) have noted, a key feature that distinguishes European style contracts from union contracts in the U.S. is that most workers receive an idiosyncratic wage cushion that “tops up” their wage over the contractual wage floor. We therefore adopt some of the methods that have

been developed to study the effect of minimum wages —specifically models of *wage spillovers*— to the study of sectoral wage floors. We also extend the seminal approach of DiNardo, Fortin and Lemieux (1996) to develop a series of counterfactuals that allow us to show how changes in wage floors, changes in wage cushions, and re-allocations of workers across different floor categories all contributed to wage adjustments over the past decade in Portugal.

Since wage cushions are set by the employer, rather than by the sectoral bargain itself, they introduce an important source of wage flexibility both to the cross-sectional wage distribution at a point in time, and to changes in wages for individuals and groups over time. We show that variation in wage cushions contributes significantly to many of the standard “wage gaps” in the labor market, including differences by gender, education, age, and between more and less profitable employers. The variation in wage cushions is particularly important in allowing wages to vary between more and less profitable firms covered by the same sectoral agreement, addressing a concern about sectoral bargaining that is widely raised by policy analysts (e.g., Boeri et al., 2021).

We also show that when wage floors are renegotiated in a sectoral wage bargain, only about one-half of the increase is passed through to workers’ wages. The other half is absorbed by a reduction in wage cushions. As has been well documented in the study of minimum wages, the passthrough effect of sectoral wage floors is larger for workers whose wages are closer to the floor (i.e., those with a smaller wage cushion), but in our case we find some degree of passthrough even for workers whose wages are far above the floor for their job category.

We find little evidence that employers adjust to rising wage floors by cutting overall employment: looking across 16 demographic subgroups we find one significantly negative estimate of the effect of wage floors on employment, counterbalanced by three significantly positive estimates. Moreover, the estimated employment effects of wage floors for different demographic groups are positively correlated with the estimated wage effects – the opposite of the pattern expected if wage floors reduce employment. The absence of systematic employment effects may not be too surprising in a setting where the majority of workers are receiving an employer-determined wage cushion that places their wage above the floor: a growing body of evidence suggests that when wages are set by employers, the effect of minimum wage increases is small.

Our counterfactual analysis of wage changes from before to after the recent financial crisis in Portugal shows that the remarkable declines in real wages for many groups were accomplished by a combination of declining real wage floors, declining real wage cushions, and a re-allocation of workers across wage floor categories. The re-allocation effect was particularly important for higher-educated groups, who entered new jobs at lower floors than would have been expected prior to the crisis and were also promoted less quickly to higher wage floor categories. A growing body of work summarized in Moscarini and Postel-Vinay (2018) suggests that a key characteristic of recessions is the absence of job openings at higher-paying firms. The pattern of re-allocations and slower promotions to higher wage floors observed in our sample period is very consistent with that view.

An important limitation of our study is that we only have data for one country. It is possible that some of the flexibility we document in the Portuguese labor market is absent in other labor markets. Indeed, the Portuguese labor market has long been characterized by relatively high levels of wage inequality. In other countries, institutional or legal restrictions may make it impossible for firms to reduce wage cushions when sectoral wage floors are increased. Providing evidence on how floors and cushions interact in other countries would clearly be helpful for future policy-making.

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Table 1: Characteristics of New Contracts in BTE, Workers in QP, and Merged BTE-QP Sample

Renegotiated Contracts Reported in BTE (non-duplicates, in scope)			Full Time Workers in QP (Private Sector Age 18-64)			Matched BTE-QP Sample		
Year	Number	Mean Number	Number of	Percent	Percent	Number of	Percent of	Number of Floors (9)
	Contracts	Months	Workers in	Covered	Covered	Workers with	All Workers	
	in BTE (1)	Since Last Contract (3)	QP (4)	by CBA (5)	by Sectoral CBA (6)	Assigned Floors in QP (7)	(7)/(4) (8)	
2008	192	20	1,966,522	90.0	75.0	634,300	32.3	1,935
2009	165	18	1,893,484	89.9	74.2	804,653	42.5	2,211
2010	140	20	1,897,345	91.4	75.2	835,011	44.0	2,357
2011	111	20	1,868,715	90.9	74.7	817,703	43.8	2,461
2012	50	25	1,768,599	89.1	71.6	832,861	47.1	2,566
2013	54	22	1,748,831	88.6	70.8	815,606	46.6	2,585
2014	83	26	1,778,271	88.4	70.6	825,698	46.4	2,619
2015	90	37	1,831,708	88.0	70.3	844,830	46.1	2,603
2016	103	29	1,884,758	87.0	69.6	855,602	45.4	2,641
All	988	23	16,638,233	89.3	72.5	7,266,264	43.7	21,978

Notes: CBA denotes a collective bargaining agreement. Columns 1-3 are based on non-duplicated counts of contracts for workers in mainland Portugal excluding agriculture and fisheries. Workers in QP include full-time private sector employees age 18-64. We also exclude apprentices, employees in agriculture and fisheries, employees of accounting firms, and individuals with missing data on monthly wages, education, or date of hire. Workers assigned a wage floor exclude those whose wage floor depends on firm or worker characteristics that we cannot trace, workers covered by contracts that were not renegotiated over the analysis period, and other groups (see text).

Table 2: Comparisons of Workers by CBA Coverage, and of Covered Workers by Assigned Floor Status

	All (1)	By CBA Coverage		By Floor Assignment (if covered)	
		Covered (2)	Not Covered (3)	Floor Assigned (4)	No Floor (5)
Fraction female	0.452	0.448	0.476	0.420	0.476
Fraction with high school	0.242	0.240	0.256	0.233	0.248
Fraction with university	0.191	0.169	0.376	0.156	0.181
Mean years experience	23.85	24.32	19.98	24.50	24.14
Mean tenure current job	8.34	8.59	6.31	8.69	8.48
Mean log monthly base wage (standard deviation)	6.696 (0.509)	6.675 (0.495)	6.858 (0.590)	6.664 (0.491)	6.686 (0.499)
Mean log monthly total wage (standard deviation)	6.856 (0.532)	6.837 (0.522)	7.014 (0.586)	6.837 (0.517)	6.838 (0.528)
Number person-years	16,638,233	14,852,805	1,785,428	7,266,264	7,586,541

Source: Quadros de Pessoal. Sample includes wage earners age 18-64 with non-missing base wage, education and date of hire, working under full time contract. Apprentices, workers in Madeira and the Azores, and those in agriculture and fishing are excluded.

Table 3: Proximate Contributions of Wage Floors, Cushions, and Supplements to Level and Variance of Wages

	Decomposition of Means:					Decomposition of Variances:				
	Mean Log	Mean	Relative Wage	Mean Wage	Mean	Var. Log	Var. Rel.	Var.	2 × Cov [Rel.	
	Total Wage (1)	Rel to Min. (2)	Floor (3)	Cushion (4)	Supplements (5)	Total Wage (6)	Wage Flr. (7)	Cushion (8)	Suppl. (9)	Flr. & Cush. (10)
All Workers (Percent of Total)	6.84	0.61	0.24 (40.2)	0.19 (31.4)	0.17 (28.4)	0.267	0.086 (32.3)	0.112 (42.1)	0.024 (9.1)	0.042 (15.6)
By Gender:										
Males (58.0% of obs.)	6.91	0.68	0.26 (39.1)	0.23 (34.0)	0.18 (26.8)	0.286	0.091 (31.9)	0.129 (45.2)	0.030 (10.3)	0.039 (13.5)
Females (42.0% of obs.)	6.74	0.51	0.22 (42.3)	0.14 (26.5)	0.16 (31.2)	0.226	0.078 (34.7)	0.084 (37.3)	0.017 (7.6)	0.040 (17.8)
Gender Gap (Percent of Gap)	0.16	0.16	0.05 (29.7)	0.09 (57.9)	0.02 (13.3)	0.060	0.013 (21.8)	0.045 (74.7)	0.012 (20.6)	-0.010 (-16.6)
By Education:										
<High School (61.1% of obs.)	6.65	0.43	0.14 (32.7)	0.12 (28.8)	0.16 (38.5)	0.123	0.031 (25.2)	0.061 (49.4)	0.021 (16.7)	0.010 (8.2)
High School (23.3% of obs.)	6.93	0.70	0.30 (42.7)	0.21 (29.7)	0.19 (27.6)	0.256	0.091 (35.4)	0.113 (43.9)	0.027 (10.5)	0.023 (8.8)
University (15.6% of obs.)	7.41	1.18	0.57 (48.7)	0.43 (36.6)	0.17 (14.7)	0.372	0.141 (37.8)	0.239 (64.3)	0.035 (9.4)	-0.023 (-6.3)
HS/<HS Gap (Percent of Gap)	0.28	0.27	0.16 (58.3)	0.08 (31.2)	0.03 (10.5)	0.134	0.060 (44.8)	0.052 (38.9)	0.006 (4.8)	0.012 (9.3)
University/HS Gap (Percent of Gap)	0.48	0.48	0.28 (57.4)	0.22 (46.3)	-0.02 (-4.0)	0.115	0.050 (43.1)	0.126 (109.6)	0.008 (7.1)	-0.046 (-39.7)
By Quartile of Firm VA/Worker:										
1st Quartile (25.0% of obs.)	6.51	0.28	0.11 (38.4)	0.04 (15.3)	0.13 (46.3)	0.076	0.027 (35.1)	0.044 (57.5)	0.011 (13.9)	-0.002 (-2.3)
4th Quartile (25.0% of obs.)	7.19	0.96	0.37 (38.7)	0.41 (42.2)	0.18 (19.1)	0.321	0.114 (35.6)	0.183 (57.2)	0.034 (10.5)	0.043 (13.4)
4th-1st Quartile (Percent of Gap)	0.69	0.68	0.27 (38.9)	0.36 (53.1)	0.05 (8.0)	0.245	0.088 (35.7)	0.140 (57.1)	0.023 (9.4)	0.045 (18.3)

Notes: see note to Table 2. Information on value added per worker is only available for non-financial firms. See text.

Table 4: Estimated Models for Wage Floor, Cushion and Base Wage within CBA's

CBA Fixed Effects	Log Floor		Wage Cushion		Log Base Wage		Cushion Share *	
	No	Yes	No	Yes	No	Yes	No	Yes
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Female	-0.054 (0.003)	-0.053 (0.001)	-0.086 (0.003)	-0.094 (0.002)	-0.140 (0.003)	-0.147 (0.003)	0.615	0.641
Education (yrs.)	0.031 (0.001)	0.024 (0.000)	0.027 (0.001)	0.030 (0.001)	0.058 (0.001)	0.054 (0.001)	0.457	0.560
Age	0.014 (0.000)	0.013 (0.000)	0.024 (0.000)	0.023 (0.000)	0.037 (0.000)	0.036 (0.000)	0.627	0.635
Age ² / 100	-0.010 (0.001)	-0.010 (0.001)	-0.019 (0.001)	-0.018 (0.001)	-0.029 (0.001)	-0.028 (0.001)	0.650	0.631
<i>Indicators for Decile of Mean Log Value Added per Worker at Firm:</i>								
Decile 2	0.001 (0.004)	-0.004 (0.002)	0.016 (0.003)	0.014 (0.003)	0.017 (0.005)	0.010 (0.004)	0.934	1.434
Decile 3	0.031 (0.008)	-0.001 (0.002)	0.021 (0.006)	0.044 (0.003)	0.052 (0.005)	0.043 (0.004)	0.399	1.016
Decile 4	0.047 (0.012)	0.000 (0.004)	0.034 (0.012)	0.073 (0.005)	0.081 (0.011)	0.074 (0.008)	0.416	0.994
Decile 5	0.030 (0.008)	0.009 (0.003)	0.077 (0.006)	0.097 (0.004)	0.107 (0.005)	0.106 (0.005)	0.718	0.913
Decile 6	0.037 (0.007)	0.006 (0.003)	0.115 (0.005)	0.137 (0.005)	0.152 (0.007)	0.143 (0.005)	0.754	0.955
Decile 7	0.042 (0.004)	0.016 (0.002)	0.155 (0.005)	0.174 (0.005)	0.197 (0.006)	0.191 (0.006)	0.787	0.915
Decile 8	0.079 (0.007)	0.028 (0.003)	0.188 (0.009)	0.223 (0.008)	0.268 (0.009)	0.251 (0.009)	0.704	0.887
Decile 9	0.114 (0.009)	0.039 (0.007)	0.246 (0.010)	0.290 (0.010)	0.360 (0.012)	0.329 (0.014)	0.684	0.882
Decile 10	0.234 (0.023)	0.052 (0.004)	0.336 (0.016)	0.414 (0.011)	0.570 (0.019)	0.466 (0.012)	0.589	0.888
R-squared	0.387	0.656	0.294	0.353	0.499	0.536	--	--

Notes: Models are estimated on 6,518,290 person-year observations for workers covered by collective bargaining agreements (CBA's) at firms with non-missing value-added data. All models also include year effects. Standard errors, clustered by firm, in parentheses.

* Share of the effect of covariate in row heading on base wage that is attributable to effect on wage cushion.

Table 5: Alternative Models for Renegotiated Wage Floors

<i><u>Explanatory variables (degrees of freedom)</u></i>	Adjusted R-squared
1. Year effects (7)	0.787
2. Year effects (7) and modal industry effects (15)	0.834
3. Year effects (7), modal industry effects (15), and worker characteristics (3)	0.838
4. Year effects (7), modal industry effects (15), worker characteristics (3), and dummies for elapsed time since last renegotiation (6)	0.853
5. Year × modal industry effects (71), worker characteristics (3), and dummies for elapsed time since last renegotiation (6)	0.898
6. CBA effects (454)	0.981
7. Contract effects (454) and worker characteristics (3)	0.982

Notes: dependent variable is change in real wage floor for a specific floor group within a collective bargaining agreement (CBA). Estimates are weighted by the number of workers covered by the floor. Sample includes 5,358 renegotiated floors that were set strictly above the minimum wage at the current and most recent previous renegotiation.

Table 6: Models for Change in Average Wage Floor in Renegotiated Collective Bargaining Agreements

	Measure of Distribution of Real Value Added per Worker Used:					
	Mean (1)	10th Pctile (2)	25th Pctile (3)	50th Pctile (4)	75th Pctile (5)	90th Pctile (6)
Change in Real Value Added/Worker	0.068 (0.014)	0.007 (0.005)	0.037 (0.011)	0.067 (0.015)	0.034 (0.011)	0.032 (0.007)
<u>Other Controls:</u>						
Cumulative Inflation since last renegotiation	0.202 (0.086)	0.032 (0.089)	0.034 (0.074)	0.142 (0.068)	0.115 (0.071)	0.040 (0.071)
Share of Females Covered by CBA	0.015 (0.003)	0.017 (0.004)	0.015 (0.004)	0.014 (0.003)	0.016 (0.003)	0.015 (0.003)
Share of Univ. Grads Covered by CBA	-0.002 (0.006)	-0.008 (0.006)	-0.008 (0.006)	-0.007 (0.005)	-0.006 (0.006)	-0.005 (0.006)
Mean Age of Workers Covered by CBA	0.000 (0.001)	0.000 (0.001)	0.000 (0.001)	0.000 (0.001)	0.000 (0.001)	0.000 (0.001)
Year effects and dummies for time since last renegotiation	yes	yes	yes	yes	yes	yes
R-squared	0.946	0.931	0.933	0.943	0.934	0.934

Notes: dependent variable is change in real average wage floor in a renegotiated collective bargaining agreement (CBA). Estimates are weighted by the number of workers covered by the CBA. Standard errors (in parentheses) are clustered by CBA. Models in different columns use different summary statistics -- as indicated in the column heading -- for the distribution of changes in real value added per worker among firms covered by the CBA.

Table 7: Models for 2010-16 Change in Average Wage Floor -- Renegotiated Collective Bargaining Agreements

	Measure of Distribution of Real Value Added per Worker Used:					
	Mean	10th Pctile	25th Pctile	50th Pctile	75th Pctile	90th Pctile
	(1)	(2)	(3)	(4)	(5)	(6)
<u>No other control variables:</u>						
Change in Real Value Added/Worker (2009-2015)	0.134 (0.025)	0.105 (0.027)	0.115 (0.037)	0.131 (0.025)	0.078 (0.023)	0.055 (0.026)
R-squared	0.475	0.383	0.421	0.510	0.284	0.178
<u>With Controls for Industry:</u>						
Change in Real Value Added/Worker (2009-2015)	0.093 (0.040)	0.068 (0.029)	0.074 (0.042)	0.094 (0.036)	0.040 (0.027)	0.033 (0.025)
R-squared	0.570	0.551	0.557	0.574	0.472	0.475

Notes: dependent variable is change in real average wage floor from 2010 to 2016 in collective bargaining agreements (CBA's) that were renegotiated at least once. Estimates are weighted by the number of workers in the agreement. Robust standard errors in parentheses. Models in different columns use different summary statistics -- as indicated in the column heading -- for the distribution of changes in real value added per worker among firms covered by the CBA over the 2009-2015 interval.

Table 8: Models for Effect of Changes in Wage Floors on Changes in Real Wages of Stayers

	Models for Change in Log Base Wage of Stayers				Models for Change in Log Total Wage of Stayers			
	Individual-level wages		Firm-wide average wages		Individual-level wages		Firm-wide average wages	
	OLS	IV**	OLS	OLS	OLS	IV**	OLS	OLS
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Simulated Change in Base or Total Wage*	0.458 (0.016)	0.530 (0.020)	0.550 (0.021)	0.546 (0.031)	0.446 (0.034)	0.536 (0.044)	0.555 (0.045)	0.521 (0.049)
Change in Real Value-added per Worker at Firm (Coeff×10)	0.021 (0.005)	0.021 (0.005)	0.021 (0.005)	0.021 (0.005)	0.017 (0.008)	0.017 (0.008)	0.017 (0.008)	0.017 (0.008)
Share of Workers with Renegotiated Floor (Coeff×10)	--	--	--	0.000 (0.005)	--	--	--	0.000 (0.012)
Share with Renegotiated Floor × Mean Simulated Change	--	--	--	0.008 (0.031)	--	--	--	0.068 (0.050)
Demographic Controls and Year Effects	yes	yes	yes	yes	yes	yes	yes	yes
First stage coefficient (instrument=mean simulated change for all workers present in previous year)	--	1.029 (0.004)	--	--	--	1.027 (0.004)	--	--
R-squared	0.092	0.092	0.229	0.229	0.030	0.030	0.078	0.078

Notes: standard errors (clustered by firm) in parentheses. Dependent variable is change in individual base wages (columns 1-2), individual total wages (columns 5-6), firm-wide average change in base wages (columns 3-4), or firm-wide average change in total wages (columns 7-8) for workers who remain at the firm from previous to current year. Models are estimated on worker-level data for 2,785,220 workers who remain at the same firm from previous to current year, but in columns 3-4 and 7-8 dependent variable and all covariates are firm-wide averages (so estimates are identical to estimates based on firm-wide average wage changes, weighing by the number of stayers). Demographic controls are shares of females and university educated workers and mean age of workers at the firm as of the previous year.

* In columns 1-2 this variable is the simulated change in the individual's base wage, based on the actual change in the real wage floor for the individual and assuming that (absolute) gap between the wage floor and the base wage remains constant. In columns 5-6 this variable is the simulated change in the individual's total wage, based on the actual change in the real wage floor for the individual and assuming that (absolute) gap between the wage floor and the total wage remains constant. In columns 3-4 (7-8) this variable is the mean simulated change in base wages (total wages) for all workers who were present in the previous year.

** Model estimated by instrumental variables, treating the simulated change in the individual's base wage (column 2) or total wage (column 6) as endogenous and using as an instrument the mean simulated change in base wages (column 2) or total wages (column 6) for all workers who were present in the previous year.

Table 9: Estimated Passthrough Rates for Floor Increases, by Subgroup

	Fraction of Stayers in Group (1)	Mean Relative Wage Floor (2)	Mean Wage Cushion (3)	Mean Wage Supplements (4)	Estimated Passthrough Rate of Wage Floor Changes	
					Base Wage (5)	Total Wage (6)
<u>Males</u>						
<High School, Age 18-24	0.011	0.074	0.078	0.183	0.50 (0.06)	0.68 (0.11)
<High School, Age 25-44	0.206	0.162	0.203	0.182	0.47 (0.03)	0.54 (0.09)
<High School, Age 45-64	0.188	0.190	0.271	0.173	0.45 (0.03)	0.46 (0.06)
High School, Age 18-24	0.007	0.108	0.095	0.201	0.39 (0.09)	0.48 (0.16)
High School, Age 25-44	0.086	0.297	0.302	0.186	0.43 (0.05)	0.42 (0.12)
High School, Age 45-64	0.031	0.429	0.536	0.163	0.31 (0.04)	0.23 (0.12)
University, Age 25-44	0.061	0.545	0.613	0.125	0.33 (0.06)	0.29 (0.09)
University, Age 45-64	0.016	0.720	0.988	0.112	0.20 (0.06)	0.24 (0.13)
<u>Females</u>						
<High School, Age 18-24	0.005	0.028	0.028	0.153	0.59 (0.04)	0.56 (0.12)
<High School, Age 25-44	0.121	0.093	0.075	0.157	0.71 (0.02)	0.63 (0.07)
<High School, Age 45-64	0.099	0.120	0.121	0.152	0.77 (0.03)	0.76 (0.05)
High School, Age 18-24	0.005	0.067	0.062	0.189	0.37 (0.13)	0.36 (0.15)
High School, Age 25-44	0.074	0.223	0.205	0.172	0.44 (0.04)	0.53 (0.09)
High School, Age 45-64	0.023	0.322	0.387	0.144	0.50 (0.04)	0.47 (0.07)
University, Age 25-44	0.057	0.442	0.477	0.129	0.31 (0.05)	0.37 (0.08)
University, Age 45-64	0.009	0.618	0.771	0.106	0.36 (0.07)	0.31 (0.10)

Notes: passthrough rates are estimated from OLS models relating firm-wide average of change in mean log base wage or mean log total wage of firm stayers to average simulated change in base wages or total wages of all workers that were present in previous year. See note to Table 8, columns 3 and 7.

Table 10: Models for Effect of Changes in Wage Floors on Change in Firm-wide Employment

	Dependent Variable = Change in Log Employment					
	All Firms			Firms Covered by Sectoral CBA's		
	(1)	(2)	(3)	(4)	(5)	(6)
Mean of Simulated Change in Total Wage of Employees (using actual Floor Changes)	0.432 (0.168)	0.165 (0.177)	0.373 (0.186)	0.412 (0.179)	0.104 (0.190)	0.317 (0.198)
Change in Real Value-added per Worker at Firm	0.027 (0.003)	0.026 (0.003)	0.026 (0.003)	0.029 (0.003)	0.028 (0.003)	0.028 (0.003)
Share of Workers with Renegotiated Floors (Coeffx10)	--	--	-0.030 (0.037)	--	--	-0.027 (0.040)
Share with Renegotiated Floor x Mean Simulated Change	--	--	-0.303 (0.305)	--	--	-0.316 (0.316)
Demographic Controls	no	yes	yes	no	yes	yes
Year Effects	yes	yes	yes	yes	yes	yes
R-squared	0.014	0.027	0.027	0.015	0.029	0.029

Notes: standard errors (clustered by firm) in parentheses. Dependent variable is change in log of employment at the firm, including all workers counted in our main QP data set. Models are weighted OLS estimates, weighted by the number of employees. Mean of simulated change in total wage (covariate in first row) is calculated by incrementing real wage floors of employees in previous year but assuming no change in other wage components. Demographic controls are shares of females and university educated workers and mean age of workers at the firm as of the previous year. In columns 4-6, sample is restricted to firms where the modal collective bargaining agreement (CBA) covering workers at the firm is a sectoral contract.

Table 11: Summary of Counterfactual Scenarios

- A. All workers in 2010, with actual 2010 floors, cushions, supplements
- B. Start with A, increment each floor by actual change 2010-16
B minus A captures floor adjustments, holding constant cushions and supplements
- C. Start with B, reweight skill groups to 2016 shares
C minus B captures demographic changes
- D. All workers in 2016, with 2016 floors, but 2010 cushions and supplements
D minus C captures the reallocation of workers across floor groups, holding constant floors, cushions, and supplements
- E. Start with D but update to actual 2016 cushions
E minus D captures adjustment of cushions within wage floor groups
- F. Start with E but update to actual 2016 supplements
(= All workers in 2016 with 2016 floors, cushions and supplements)
F minus E captures adjustment of supplements within floor-groups

Table 12: Components of Adjustment of Real Wages, 2010-2016

Mean Log Wage in 2010	Simulated Components of Real Wage Change										Share Workers w/ Floor Equal to Min Wage		
	Components of Real Wage in 2010				Change in Real Wage 2010-2016	Change in				Actual 2010 (in col 1)	After Change in Floors (in col 6)	Actual 2016	
	Rel. Floor	Cushion	Supplier's	Change in Floors		Reweighting across Skills Groups	Reallocation to New Floor Groups	Cushion within Floor Group	Supplier's within Floor Group				
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	
All	6.863	0.250	0.195	0.166	-0.017	-0.022	0.074	-0.048	-0.025	0.005	0.126	0.187	0.205
Males	6.931	0.271	0.234	0.174	-0.022	-0.028	0.077	-0.050	-0.026	0.006	0.074	0.124	0.147
Females	6.767	0.220	0.140	0.154	-0.001	-0.015	0.079	-0.045	-0.024	0.004	0.199	0.277	0.277
Educ<HS	6.677	0.142	0.122	0.160	-0.022	-0.014	0.009	-0.012	-0.007	0.002	0.165	0.233	0.259
Educ=HS	6.994	0.328	0.228	0.187	-0.126	-0.033	0.025	-0.092	-0.034	0.007	0.081	0.151	0.210
Educ=Univ.	7.491	0.611	0.468	0.160	-0.161	-0.044	0.019	-0.084	-0.062	0.010	0.015	0.037	0.063
Age 18-24	6.582	0.114	0.040	0.176	-0.009	-0.004	0.040	-0.038	-0.016	0.009	0.246	0.347	0.411
Age 25-44	6.870	0.260	0.192	0.167	-0.025	-0.024	0.071	-0.046	-0.029	0.004	0.121	0.173	0.192
Age 45-64	6.907	0.259	0.234	0.163	-0.024	-0.023	0.065	-0.051	-0.020	0.006	0.109	0.183	0.197
Males by Education and Age													
<HS, Age 18-24	6.552	0.078	0.045	0.177	-0.024	0.000	0.000	-0.018	-0.010	0.005	0.173	0.259	0.347
<HS, Age 25-44	6.723	0.154	0.145	0.172	-0.032	-0.022	0.000	-0.004	-0.007	0.002	0.094	0.148	0.186
<HS, Age 45-64	6.820	0.196	0.206	0.167	-0.060	-0.027	0.000	-0.023	-0.013	0.002	0.077	0.139	0.163
HS, Age 18-24	6.643	0.154	0.043	0.193	-0.056	-0.012	0.000	-0.040	-0.017	0.013	0.194	0.277	0.366
HS, Age 25-44	7.065	0.368	0.250	0.195	-0.177	-0.037	0.000	-0.112	-0.035	0.006	0.045	0.092	0.146
HS, Age 45-64	7.476	0.557	0.463	0.204	-0.209	-0.042	0.000	-0.132	-0.050	0.015	0.017	0.041	0.071
Univ, Age 18-24	7.043	0.490	0.154	0.146	-0.147	-0.039	0.000	-0.094	-0.041	0.027	0.037	0.076	0.114
Univ, Age 25-44	7.541	0.624	0.507	0.158	-0.181	-0.042	0.000	-0.075	-0.073	0.009	0.008	0.019	0.040
Univ, Age 45-64	8.058	0.840	0.794	0.171	-0.185	-0.043	0.000	-0.113	-0.047	0.017	0.003	0.008	0.017
Females by Education and Age													
<HS, Age 18-24	6.465	0.045	0.011	0.158	-0.014	0.012	0.000	-0.021	-0.005	0.001	0.418	0.536	0.603
<HS, Age 25-44	6.533	0.093	0.043	0.144	0.007	0.001	0.000	0.004	-0.001	0.002	0.309	0.361	0.379
<HS, Age 45-64	6.601	0.128	0.081	0.140	-0.026	-0.003	0.000	-0.020	-0.006	0.003	0.233	0.360	0.372
HS, Age 18-24	6.565	0.100	0.021	0.191	-0.048	-0.005	0.000	-0.035	-0.012	0.004	0.326	0.474	0.564
HS, Age 25-44	6.840	0.255	0.159	0.174	-0.118	-0.031	0.000	-0.061	-0.029	0.003	0.099	0.195	0.258
HS, Age 45-64	7.152	0.391	0.337	0.171	-0.185	-0.039	0.000	-0.112	-0.041	0.007	0.038	0.092	0.153
Univ, Age 18-24	6.956	0.440	0.115	0.148	-0.156	-0.048	0.000	-0.083	-0.046	0.021	0.094	0.158	0.208
Univ, Age 25-44	7.317	0.538	0.367	0.160	-0.175	-0.045	0.000	-0.073	-0.063	0.007	0.022	0.056	0.090
Univ, Age 45-64	7.765	0.763	0.583	0.167	-0.203	-0.046	0.000	-0.131	-0.040	0.015	0.009	0.025	0.043

Notes: Wages refer to real monthly wages, including regular supplementary payments. Column (11) reports the actual share of workers at a floor equal to the minimum wage in 2010. Column (12) considers the updating in collective bargaining floors, ceteris paribus, reporting the share of workers who would have had a floor at the minimum wage in 2016 if they had stayed in their same floor category as in 2010. Column (13) reports the actual share of workers at a floor equal to the minimum wage in 2016. The change from columns (12) to (13) hence reflects demographic shifts in the economy and re-allocation of workers with a given age-gender-education across floors. See text for description of simulations for decomposition.

ONLINE APPENDIX: WAGE FLEXIBILITY UNDER SECTORAL BARGAINING

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Appendix A: Assigning Wage Floors from BTE to Workers in QP

We assigned workers in the linked employer-employee dataset (QP) to wage floors published in the Labor Bulletin (*Boletim do Trabalho e Emprego*, BTE). Appendix Table A.1 reports the constraints on the collective bargaining agreements (CBA's) published in BTE that we imposed to define our analysis sample. In turn, Appendix Table A.2 reports the constraints we imposed on the QP dataset.

A total of 1,467 contracts including wage clauses were published in BTE between January 2008 and December 2016. Parallel agreements were signed by different trade unions with the same employer(s), presenting exactly the same contents. We identified 406 such redundant contracts, leaving 1,061 “consolidated” CBA's eligible for analysis. Fifty contracts fall outside the set of industries under study, whereas 23 fall outside the geographic scope of the analysis. Hence we consider 988 non-duplicate contracts within the scope of our analysis.

However, a few contracts (two) were never enforced over the period October 2008 to 2016; another 22 agreements do not appear to have any covered workers in QP in the 2008-2016 period. An additional 267 contracts defined wage floors conditional on information that cannot be identified in QP. In some cases this information pertains the worker (such as academic grades, subjective evaluations of CV or performance, type of schedule, or her progression along a set of occupational “steps”); in other cases this information pertains to the firm (such as the category of the establishment in accommodation, food and leisure services, or the average corporate income tax paid in the recent past). Therefore, the analysis set includes 697 collective bargaining agreements.

We restricted the QP dataset to wage-earners aged 18 to 64, with non-missing base wage, education and date of hire, reported working full-time; we also excluded agriculture and fisheries, those working in Madeira and the Azores, apprentices, and workers in accounting firms. These constraints resulted in a dataset of 16.6 million

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observations worker-year. Eleven percent of these workers were reported not covered by a CBA. Another 7% were reported as covered by a CBA but the job category was not specified. For 10%, no renegotiation of the collective bargaining contract took place between 2008 and 2016; and for a residual 1% a floor update was identified, but its dates of enforcement fell outside our analysis period.

For approximately one fourth of the workforce, we were unable to assigned a wage floor, either because: we could not find the worker's floor group in BTE, even though we found her contract (3% of the observations); or the identified wage floor was actually enforced retrospectively and the wage floor at the time of QP was unknown (~1% of observations); or the wage floor depended on information that could not be identified in QP. Hence our analysis sample includes 7.3 million observation worker-year.

Appendix B: Dynamics of CBA coverage

In a given year, individuals can be classified into 3 states: c = employed in covered job; n = employed in uncovered job; o = out of work (i.e., not in QP). Let $N^{ij}(t)$ represent the number of workers who move from state i to state j between $t - 1$ and t , let $N^j(t)$ represent the number in state j in period t (aggregating across all previous states) and let $N^{i\cdot}(t - 1)$ represent the number in state i in year $t - 1$ (aggregating across all subsequent states). Finally, let $E(t) = N^c(t) + N^n(t)$ represent the number of employed workers in year t . Then the coverage rate in year t is:

$$\begin{aligned}
 C(t) &\equiv \frac{N^c(t)}{E(t)} \\
 &= \frac{N^{nc}(t) + N^{cc}(t) + N^{oc}(t)}{E(t)} \\
 &= \frac{N^{nc}(t)}{N^{nc}(t) + N^{nn}(t)} \times \frac{N^{nc}(t) + N^{nn}(t)}{E(t)} \\
 &\quad + \frac{N^{cc}(t)}{N^{cc}(t) + N^{cn}(t)} \times \frac{N^{cc}(t) + N^{cn}(t)}{E(t)} \\
 &\quad + \frac{N^{oc}(t)}{N^{oc}(t) + N^{on}(t)} \times \frac{N^{oc}(t) + N^{on}(t)}{E(t)}
 \end{aligned}$$

Now let

$$\delta^n(t - 1) = \frac{N^{no}(t)}{N^{n\cdot}(t - 1)}$$

represent the fraction of noncovered workers in year $t - 1$ who are out of employment in year t , and similarly let

$$\delta^c(t - 1) = \frac{N^{co}(t)}{N^c(t - 1)}$$

represent the fraction of covered workers in year $t - 1$ who are out of employment in year t . Then we can write:

$$\begin{aligned}\frac{N^{nc}(t) + N^{nn}(t)}{E(t)} &= \frac{N^{nc}(t-1)(1 - \delta^n(t-1))}{E(t-1)} \times \frac{E(t-1)}{E(t)} \\ &= \frac{(1 - C(t-1))(1 - \delta^n(t-1))}{E(t)/E(t-1)} \\ \frac{N^{cc}(t) + N^{cn}(t)}{E(t)} &= \frac{N^{cc}(t-1)(1 - \delta^c(t-1))}{E(t-1)} \times \frac{E(t-1)}{E(t)} \\ &= \frac{C(t-1)(1 - \delta^c(t-1))}{E(t)/E(t-1)}\end{aligned}$$

Finally, let

$$\mu(t) = \frac{N^{oc}(t) + N^{on}(t)}{E(t)}$$

represent the fraction of the employed workforce in period t who were not employed in $t - 1$.

Then

$$C(t) = \lambda^{nc}(t)(1 - C(t-1))w_{1t} + \lambda^{cc}(t)C(t-1)w_{2t} + \lambda^{oc}(t)\mu(t) \quad (\text{B.1})$$

where

$$\begin{aligned}\lambda^{nc}(t) &= \frac{N^{nc}(t)}{N^{nc}(t) + N^{nn}(t)} \\ \lambda^{cc}(t) &= \frac{N^{cc}(t)}{N^{cc}(t) + N^{uc}(t)} \\ \lambda^{oc}(t) &= \frac{N^{oc}(t)}{N^{oc}(t) + N^{on}(t)}, \\ w_{1t} &= \frac{(1 - \delta^n(t-1))}{E(t)/E(t-1)} \\ w_{2t} &= \frac{(1 - \delta^c(t-1))}{E(t)/E(t-1)}.\end{aligned}$$

The term λ^{nc} is the transition rate from an uncovered job in $t - 1$ to a covered job in t , while λ^{cc} is the retention rate from a covered job in $t - 1$ to a covered job in t . The terms w_1 and w_2 are adjustment factors that account for transitions out of work, and for overall employment growth between periods. Note that if there is no flow of workers in or out of employment between periods then $w_{1t} = w_{2t} = 1$, and $\mu(t) = 0$, so equation (B.1) becomes the simple flow equation:

$$C(t) = \lambda^{nc}(t)(1 - C(t-1)) + \lambda^{cc}(t)C(t-1).$$

Appendix Table B.1 shows employment counts in the current and past year, the transition rates $\lambda^{nc}(t)$, $\lambda^{cc}(t)$, $\lambda^{oc}(t)$, the attrition rates $\delta^n(t-1)$, $\delta^c(t-1)$, the share of current employees who were not working last year, $\mu(t)$, and the adjustment factors w_{1t} , w_{2t} for equation (B.1).

Appendix Table A1: Collective Bargaining Agreements BTE,
Constraints on Analysis of Wage Floors, 2008-2016.

	Number of Agreements (1)	Percent of Agreements (2)
Population BTE [*]	1,467	100
Of which:		
duplicate agreement	406	-28
non-duplicate agreement	1,061	72
Outside scope of analysis:		
industry agriculture, fisheries	-50	-3
location Azores or Madeira	-23	-2
Outside analysis set:		
contract not reported in QP	-22	-1
wage table not enforced Oct. 2008 - 2016	-2	0
wage floor depends on info. not reported in QP	-267	-18
Analysis set of agreements from BTE	697	48

Source: Portugal, MTSS, Labor Bulletin, 2008-2016.

* Collective agreements published 2008-2016 that set salary scales (new or updated). A duplicate agreement replicates another one signed by the same employers but a different trade union. Some agreements define sets of sub-floors along dimension(s) not observed in QP (e.g., worker attributes such as academic grades, subjective evaluations of CV or performance; firm attributes such as category of establishment in accommodation, food and leisure services, or the average corporate income tax paid in the recent past).

Appendix Table A2: Workers in QP, Analysis Set, 2008-2016.

	Number of Worker-Year Obs. (1)	Percent (2)
Population full-time wage-earners QP [*] :	16,638,233	100
Not covered by collective bargaining agreement (CBA):	1,785,428	11
Covered by CBA but no wage floor:		
residual/unspecified CBA category	1,179,189	7
CBA never updated 2008-2016	1,678,799	10
CBA updated but floors in 2008-2016 unaffected	150,586	1
Covered by CBA but could not assign wage floor:		
wage floor depends on information not reported in QP	3,845,184	23
could not find floor category in BTE	538,512	3
back-dated, no information on earlier floor	194,271	1
Successfully assigned CBA and wage floor	7,266,264	44

Source: Portugal, MTSS, Labor Bulletin, 2008-2016 and Portugal, MTSS, QP, 2008-2016.

^{*}Wage-earners aged 18 to 64, with non-missing base wage, education and seniority within the firm, reported on full-time contract; excludes agriculture and fisheries, Madeira and the Azores, and apprentices. Some agreements define sets of sub-floors along dimension(s) not observed in QP (e.g., worker attributes such as academic grades, subjective evaluations of CV or performance; firm attributes such as category of establishment in accommodation, food and leisure services, or the average corporate income tax paid in the recent past). "Could not find match to category in BTE" refers to specific categories that could not be matched to a salary group in BTE, even though the agreement was matched. Wage floors represent floors known to the employer at the QP reference date and do not represent retroactively imposed floors.

Appendix Table A3: New Contracts in BTE -- Total and Adjusting for Duplicates and Scope

Year	New Contracts Reported in BTE				New Contracts Reported in BTE (non-duplicates, in scope)				
	Number Contracts in BTE (1)	Number of CBA's by Type:			Number of Non- duplicated Contracts (5)	Ratio: column 5 to column 1 (%) (6)	Number of CBA's by Type:		
		Sectoral (2)	Multi-Firm (3)	Single Firm (4)			Sectoral (7)	Multi-Firm (8)	Single Firm (9)
2008	293	172	25	96	192	65.5	113	16	63
2009	246	141	21	82	165	67.1	88	16	59
2010	216	140	24	50	140	64.8	86	16	36
2011	167	93	22	51	111	66.5	58	14	38
2012	70	30	7	32	50	71.4	19	4	26
2013	78	25	16	37	54	69.2	17	10	27
2014	126	43	20	63	83	65.9	29	10	44
2015	131	66	15	49	90	68.7	43	9	37
2016	140	67	20	53	103	73.6	48	14	41
All	1,467	777	170	513	988	67.3	501	109	371

Notes: CBA denotes a collective bargaining agreement. Columns 1-4 are based on reported CBA's in BTE; columns 5-9 are non-duplicated counts of contracts for workers in mainland Portugal excluding agriculture and fisheries.

Appendix Table B1: Dynamics of Collective Bargaining Coverage

	All Years Combined			By Year, Pooling Males and Females									
	All	Males	Females	2009	2010	2011	2012	2013	2014	2015	2016		
<u>Employment counts</u>													
E(t)	14,671,711	8,010,391	6,661,320	1,893,484	1,897,345	1,868,715	1,768,599	1,748,831	1,778,271	1,831,708	1,884,758		
E(t-1)	14,753,475	8,110,959	6,642,516	1,966,522	1,893,484	1,897,345	1,868,715	1,768,599	1,748,831	1,778,271	1,831,708		
E(t)/E(t-1)	0.994	0.988	1.003	0.963	1.002	0.985	0.946	0.989	1.017	1.030	1.029		
<u>Transition Rates to Covered Jobs (conditional on employment)</u>													
λ^{nc} (uncovered-covered)	0.103	0.114	0.091	0.126	0.273	0.074	0.064	0.077	0.069	0.083	0.086		
λ^{cc} (covered-covered)	0.986	0.986	0.987	0.983	0.987	0.993	0.979	0.988	0.991	0.990	0.981		
λ^{nc} (out of work-covered)	0.881	0.883	0.878	0.883	0.896	0.900	0.884	0.882	0.880	0.866	0.856		
<u>Attrition Rates (probabilities of transitioning to out-of-QP)</u>													
δ^n (loss from uncovered)	0.222	0.211	0.235	0.294	0.322	0.193	0.197	0.186	0.194	0.188	0.201		
δ^c (loss from covered)	0.217	0.211	0.223	0.233	0.255	0.218	0.231	0.204	0.193	0.197	0.197		
<u>Share of Employment in t that was out-of-QP in t-1</u>													
μ	0.213	0.201	0.227	0.209	0.263	0.204	0.184	0.193	0.206	0.220	0.220		
<u>Adjustment Factors to account for out-of-QP</u>													
$w_1(t)$	0.782	0.799	0.763	0.734	0.677	0.819	0.848	0.823	0.793	0.788	0.777		
$w_2(t)$	0.788	0.799	0.774	0.797	0.744	0.794	0.812	0.805	0.794	0.779	0.781		
<u>Three components of equation (1)</u>													
$\lambda^{nc}(t)(1-C(t-1))w_1(t)$	0.008	0.009	0.008	0.009	0.019	0.005	0.005	0.007	0.006	0.008	0.008		
$\lambda^{cc}(t)C(t-1)w_2(t)$	0.696	0.709	0.680	0.705	0.660	0.720	0.723	0.709	0.696	0.682	0.674		
$\lambda^{oc}(t)\mu(t)$	0.188	0.178	0.199	0.185	0.236	0.184	0.163	0.170	0.182	0.190	0.188		
Sum (=coverage rate)	0.892	0.896	0.887	0.899	0.914	0.909	0.891	0.886	0.884	0.880	0.870		

Source: Sample is the same as in Table 1, column 4, years 2009-2016. See notes to that table.

Appendix Table C1: Comparisons of Workers Covered by CBA's with an Assigned Floor to All Covered Workers

	2008	2009	2010	2011	2012	2013	2014	2015	2016
Share with floor	0.36	0.47	0.48	0.48	0.53	0.53	0.53	0.52	0.52
Frac. female with floor	0.40	0.40	0.41	0.42	0.42	0.43	0.43	0.43	0.43
Frac. female (all)	0.43	0.43	0.44	0.45	0.46	0.46	0.46	0.46	0.46
Gap in fraction female	-0.03	-0.03	-0.03	-0.02	-0.03	-0.03	-0.03	-0.03	-0.03
Mean education with floor	7.98	8.85	9.17	9.40	9.55	9.70	9.84	10.03	10.21
Mean education (all)	8.86	9.10	9.39	9.58	9.78	9.94	10.08	10.27	10.46
Gap in mean education	-0.88	-0.26	-0.22	-0.18	-0.23	-0.24	-0.24	-0.25	-0.25
Mean age with floor	38.68	38.95	39.23	39.51	39.89	40.30	40.67	40.90	41.11
Mean age (all)	38.82	39.17	39.41	39.70	40.04	40.42	40.75	40.96	41.13
Gap in mean age	-0.14	-0.22	-0.19	-0.19	-0.15	-0.11	-0.08	-0.07	-0.02
Mean tenure with floor	7.31	7.87	8.20	8.50	9.06	9.31	9.33	9.23	9.08
Mean tenure (all)	7.75	8.03	8.26	8.50	8.91	9.10	9.10	8.98	8.83
Gap in mean tenure	-0.44	-0.15	-0.06	-0.01	0.15	0.21	0.23	0.25	0.25
Mean log base wage with floor	6.58	6.68	6.70	6.68	6.66	6.66	6.67	6.67	6.68
Mean log base wage (all)	6.66	6.69	6.71	6.68	6.66	6.66	6.67	6.67	6.68
Gap in mean log base wage	-0.08	-0.02	-0.01	-0.01	0.00	0.00	0.00	0.00	0.00
Mean log total wage with floor	6.73	6.84	6.87	6.85	6.84	6.84	6.85	6.84	6.85
Mean log total wage (all)	6.81	6.84	6.87	6.85	6.83	6.83	6.83	6.83	6.84
Gap in mean log total wage	-0.08	-0.01	0.00	0.01	0.01	0.01	0.01	0.01	0.01

Source: Quadros de Pessoal. Sample includes wage earners age 18-64 with non-missing base wage, education and date of hire, working under full time contract. Apprentices, workers in Madeira and the Azores, and those in agriculture and fishing are excluded. All refers to all workers covered by collective bargaining.

Appendix Table D1: Models for 2010-16 Change in Average Wage Floor -- Renegotiated CBA's

	Extra Measure of Distribution of Real Value Added per Worker Used:				
	None (1)	10th Pctile (2)	25th Pctile (3)	75th Pctile (4)	90th Pctile (5)
<u>Without industry controls:</u>					
Change in Median Real Value Added/Worker (2009-2015)	0.131 (0.025)	0.101 (0.028)	0.101 (0.041)	0.121 (0.034)	0.120 (0.027)
Change in other Quantile of Real Value Added/Worker (2009-2015)	--	0.038 (0.021)	0.035 (0.050)	0.012 (0.022)	0.019 (0.025)
R-squared	0.510	0.534	0.523	0.513	0.526
<u>With Controls for Industry:</u>					
Change in Real Value Added/Worker (2009-2015)	0.094 (0.036)	0.066 (0.037)	0.064 (0.044)	0.099 (0.046)	0.087 (0.037)
Change in other Quantile of Real Value Added/Worker (2009-2015)	--	0.036 (0.024)	0.034 (0.053)	-0.008 (0.030)	0.015 (0.022)
R-squared	0.574	0.595	0.586	0.575	0.582

Notes: dependent variable is change in real average wage floor from 2010 to 2016 in collective bargaining agreements (CBA's) that were renegotiated at least once. Estimates are weighted by the number of workers in the agreement. Robust standard errors in parentheses. Models in different columns include different other quantiles of the change in real value added per worker among firms covered by the CBA -- as indicated in the column heading.

Appendix Table E1: OLS and IV Models for Effect of Wage Floor Increases on Base Wages

	Models for Individual-Level Changes in Base Wage Rate of Stayers					
	Including Wage Floors that Equal the Minimum Wage at Some Time in Sample Period			Excluding Wage Floors that Equal the Minimum Wage at Some Time in Sample Period		
	OLS	IV*	IV**	OLS	IV*	IV**
	(1)	(2)	(3)	(4)	(5)	(6)
Simulated Change in Base Wage	0.385 (0.026)	0.509 (0.038)	0.805 (0.122)	0.176 (0.043)	0.342 (0.064)	0.715 (0.147)
Change in Real Value-added per Worker at Firm (Coeff×10)	0.042 (0.009)	0.041 (0.009)	0.038 (0.009)	0.047 (0.011)	0.046 (0.011)	0.040 (0.011)
Demographic Controls and Year Effects	yes	yes	yes	yes	yes	yes
First stage coefficient (instr. in cols 2,5 = simulated change in base wage for all workers present in prev. year; in cols 3,6= lagged mean change in value added for other firms covered by CBA)	--	1.010 (0.005)	0.045 (0.004)	--	0.874 (0.012)	0.048 (0.005)
R-squared	0.101	0.100	0.098	0.085	0.085	0.084

Notes: standard errors (clustered by firm) in parentheses. Dependent variable is change in individual base wages for workers who remain at the firm from previous to current year. Models in columns 1-3 are estimated on worker-level data for 615,745 workers covered by renegotiated sectoral contracts who remain at the same firm and collective bargaining group from previous to current year, with two lagged values of mean log of value added per worker for firms covered by the CBA. Models in columns 4-6 further exclude workers covered by a wage floor that is ever equal to the minimum wage. Demographic controls are shares of females and university educated workers and mean age of workers at the firm as of the previous year.

* In columns 2 and 5 the instrument for the simulated change in the individual's base wage is the mean simulated change in base wages for all workers present at the firm in the previous year.

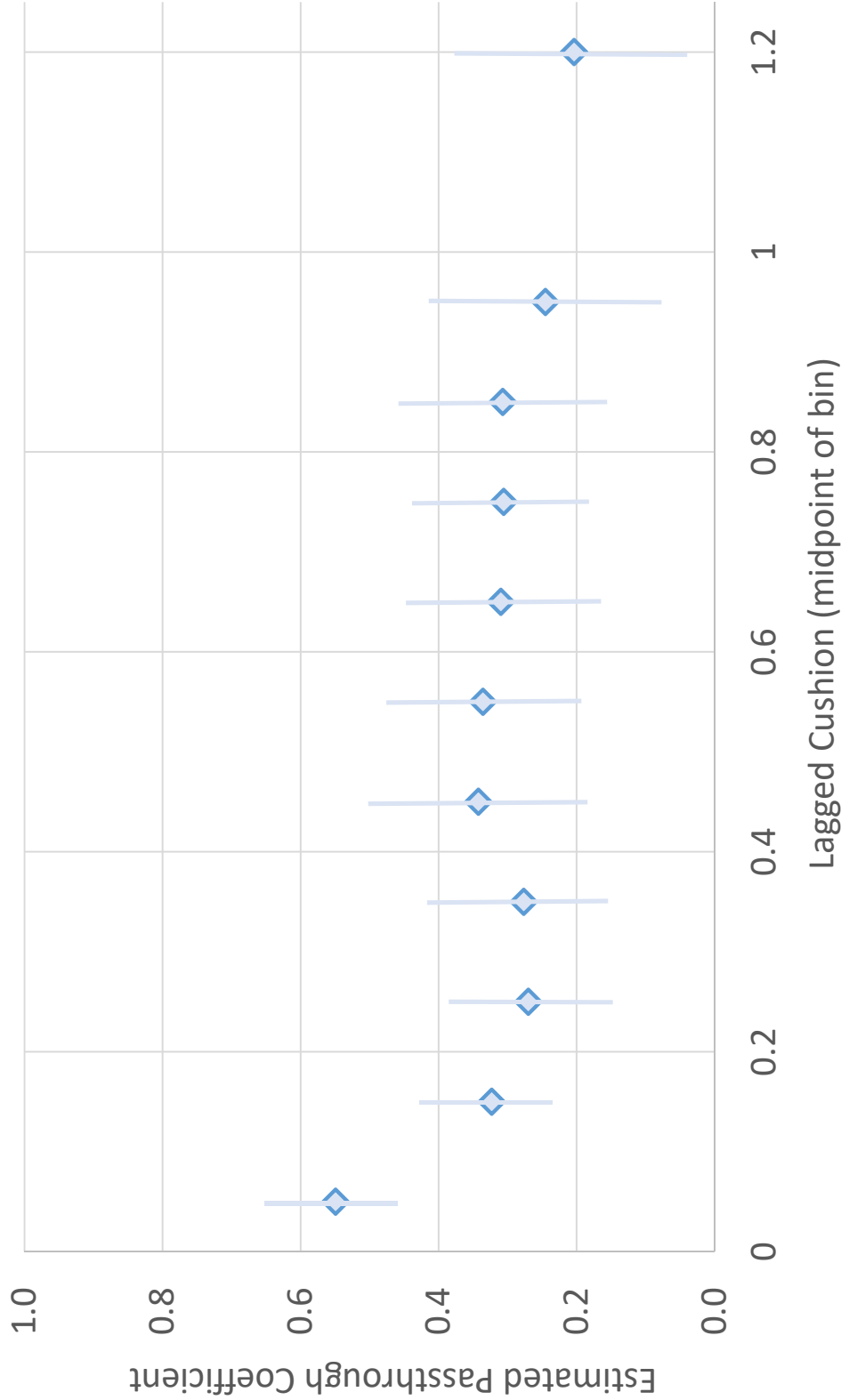
** In columns 3 and 6 the instrument for the simulated change in the individual's base wage is the lagged mean change in real value added per worker at all other firms covered by same collective bargaining agreement (CBA). Following the models in the section on the determinants of negotiated wage floors, the model includes dummies for time since contract last renegotiated and the cumulative inflation over that period.

Appendix Table E2: Estimated Wage and Employment Models by Group

	Estimated Effect of Mean Change in Wage Floor on:	
	Change in Wages (1)	Change in Empl. (2)
Males		
<HS, Age 18-24	0.68 (0.11)	1.05 (0.76)
<HS, Age 25-44	0.54 (0.09)	0.68 (0.28)
<HS, Age 45-64	0.46 (0.06)	0.88 (0.21)
HS, Age 18-24	0.48 (0.16)	-0.17 (0.80)
HS, Age 25-44	0.42 (0.12)	0.47 (0.39)
HS, Age 45-64	0.23 (0.12)	-0.24 (0.39)
Univ. Age 25-44	0.29 (0.09)	0.48 (0.51)
Univ. Age 45-64	0.24 (0.13)	-0.38 (0.43)
Females		
<HS, Age 18-24	0.56 (0.12)	-2.46 (1.22)
<HS, Age 25-44	0.63 (0.07)	0.28 (0.35)
<HS, Age 45-64	0.76 (0.05)	0.86 (0.29)
HS, Age 18-24	0.36 (0.15)	0.20 (1.01)
HS, Age 25-44	0.53 (0.09)	0.43 (0.29)
HS, Age 45-64	0.47 (0.07)	-0.04 (0.35)
Univ. Age 25-44	0.37 (0.08)	0.46 (0.31)
Univ. Age 45-64	0.31 (0.10)	-0.25 (0.51)

Notes: see notes to Table 8. Results in column 1 are taken from column 6 of Table 9. Results in column 2 are based on estimated model for employment changes including all wage-earners in the QP data set. Both sets of models include year effects and controls for the change in log value added per worker at the firm and for the average age of workers in the group.

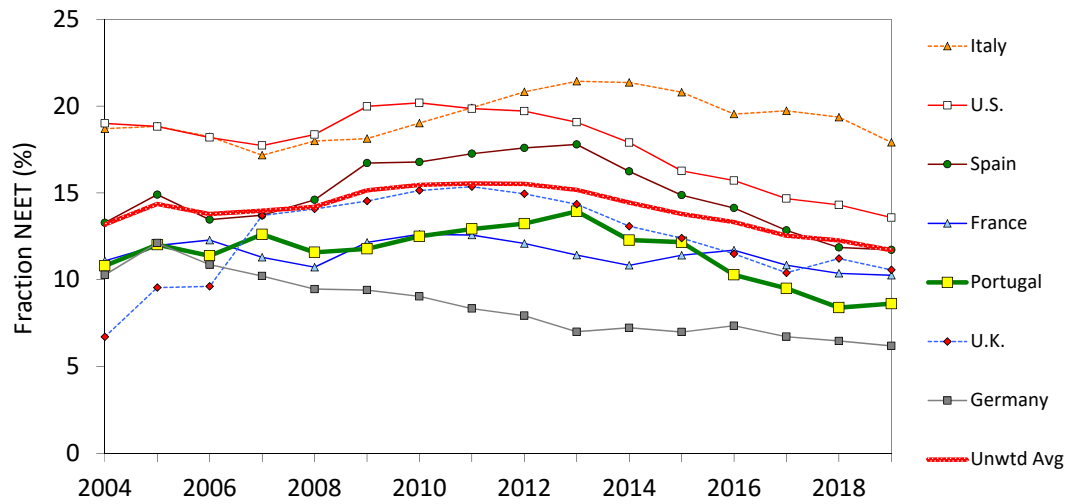
Appendix Figure E1: Estimated Passthrough Coefficients
by Level of Cushion



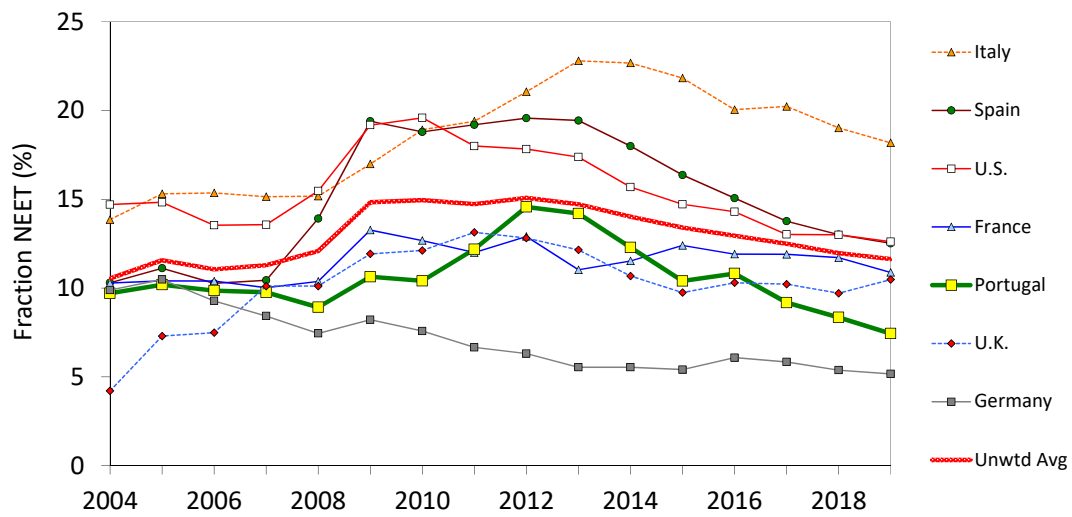
Note: figure shows point estimate and +/- 2 standard error bounds. Midpoint of highest bin is approximate.

Appendix Figure E2: Not in Employment, Education, or Training (NEET)

a) Female Youths



b) Male Youths



Source: ILO.